

Chapter 8: The Behavior of Gases

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8.1. Introduction

- ✓ Up to this point, we have made frequent use of the ideal gas to illustrate the nature of changes in the thermodynamic state of a gaseous system.
- ✓ In this chapter, the behavior of real gases is compared with ideal behavior. The differences between the two are mainly related to the atomic or molecular interactions of real gases.
- ✓ Although knowledge of the physical properties of a real gas is not required in a thermodynamic examination of the gas, an appreciation of the origin of the physical properties provides a better understanding of the thermodynamic behavior.

8.2. The P - V - T Relationships of Gases

- ✓ Experimental observation has shown that, for 1 mole of all real gases,

$$\lim_{P \rightarrow 0} \frac{PV}{RT} \rightarrow 1$$

P is the pressure of the gas

V is the molar volume of the gas

R is the universal gas constant

T is the absolute temperature of the gas

- ✓ Thus, as the pressure of the gas approaches zero, isotherms plotted on a P – V diagram approach the form of a rectangular hyperbola, given by the equation

$$PV = RT$$

8.2. The P - V - T Relationships of Gases

$$PV = RT$$

- ✓ This is the equation of state for 1 mole of an ideal gas and is called the ideal gas law.
- ✓ A gas which obeys this law over a range of states is said to behave ideally in this range of states, and a gas which obeys this law in all states is called a perfect gas.
- ✓ The perfect gas is a convenient model with which the behavior of real gases can be compared.

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8.3. Thermodynamic Properties and Mixtures of Ideal Gases

- ✓ The variation of the molar Gibbs free energy with pressure of a closed system of fixed composition and at constant temperature is found from the fundamental equation to be

$$dG = -SdT + VdP \quad \Rightarrow \quad dG = VdP$$

For 1 mole of an ideal gas, this can be written as

$$dG = \frac{RT}{P} dP = RT \ln P$$

For an isothermal change of pressure from P_1 to P_2 at T ,

$$G(P_2, T) - G(P_1, T) = RT \ln \frac{P_2}{P_1}$$

8.3. Thermodynamic Properties and Mixtures of Ideal Gases

- ✓ Since Gibbs free energies do not have absolute values (only changes in G can be measured), it is convenient to choose an arbitrary **reference state** from which the changes in Gibbs free energy can be measured.
- ✓ This reference state is called the **standard state** and is chosen as being the state of 1 mole of pure gas at 1 atm pressure and the temperature of interest.
- ✓ The Gibbs free energy of 1 mole of gas in the **standard state $G(P=1, T)$** is designated **$G^\circ(T)$** .

8.3. Thermodynamic Properties and Mixtures of Ideal Gases

- ✓ The Gibbs free energy of 1 mole of gas at any other pressure P is

$$G(P_2, T) - G(P_1, T) = RT \ln \frac{P_2}{P_1}$$



$$G(P, T) = G^0(T) + RT \ln \frac{P}{P^0}$$

This is the variation of the molar Gibbs free energy with pressure of a closed system of fixed composition and at constant temperature.

8.3.1.1 Mole Fraction

- ✓ Before discussing the thermodynamic properties of mixtures of ideal gases, it is necessary to introduce the concepts of *mole fraction*, *partial pressure* and *partial molar quantities*.
- ✓ When a system contains more than one component (i.e., when the composition of the system is variable), it is necessary to devise a means of expressing the composition.
- ✓ In this text, we use the *mole fraction*, X_i , of the component i . The mole fraction is defined as the *ratio of the number of moles of i in the system to the total number of moles of all of the components in the system*.

8.3.1.1 Mole Fraction

Mole fraction, X_i , of the component i .

For example, if the system contains n_A moles of A, n_B moles of B, and n_C moles of C, then

$$X_A = \frac{n_A}{n_A + n_B + n_C}$$

$$X_B = \frac{n_B}{n_A + n_B + n_C}$$

$$X_C = \frac{n_C}{n_A + n_B + n_C}$$

The sum of the mole fractions of all of the components in a system is unity:

$$(X_A + X_B + X_C = 1)$$

8.3.1.2 Dalton' s Law of Partial Pressures

- ✓ The contribution made to the total pressure, P , by each individual gas is called the *partial pressure* of that gas.
- ✓ The partial pressure exerted by a component gas, p_i , is thus the pressure that it would exert if it alone were present.

In a mixture of the ideal gases A, B, and C,

$$P = p_A + p_B + p_C$$

8.3.1.2 Dalton's Law of Partial Pressures

- ✓ Consider a fixed volume V' at the temperature T which contains n_A moles of an ideal gas A. The pressure exerted is thus

$$p_A = \frac{n_A RT}{V'}$$

- ✓ If n_B moles of ideal gas B are added to this constant volume containing n_A moles of gas A, the pressure increases to

$$p_A + p_B = (n_A + n_B) \frac{RT}{V'}$$

Therefore

$$\frac{p_A}{p_A + p_B} = \frac{n_A}{n_A + n_B}$$

8.3.1.2 Dalton's Law of Partial Pressures

$$\frac{p_A}{p_A + p_B} = \frac{n_A}{n_A + n_B}$$

For the gas A in the mixture, one has:

$$\frac{p_A}{P} = X_A \quad \Rightarrow \quad p_A = X_A P$$

Dalton's law of partial pressures

Thus, in a mixture of ideal gases, the partial pressure of a component gas is the product of its mole fraction and the total pressure of the gas mixture.

8.3.1.3 Partial Molar Quantities

- ✓ The partial molar value of an extensive thermodynamic variable $\bar{Q}'$ is the rate of change of Q' with respect to the addition of n_i moles of component i at constant temperature, pressure, and composition. This can be written as

$$\bar{Q}_i = \left(\frac{\partial Q'}{\partial n_i} \right)_{T, P, n_j, n_k, \dots}$$

Here Q' is the value of the extensive thermodynamic variable for an arbitrary quantity of the mixture.

Intensive & extensive variable:

Intensive: not depend on the volume of the system

Extensive: depend on the volume of the system

8.3.1.3 Partial Molar Quantities

$$\bar{Q}_i = \left(\frac{\partial Q'}{\partial n_i} \right)_{T, P, n_j, n_k, \dots}$$

- ✓ If 1 mole of i is added, at constant temperature and pressure, to a quantity of a solution which is sufficiently large that the addition causes virtually no change in the composition of the solution, the consequent increase in the value of Q' equals the value of $\bar{Q}_i$ in the solution.
- ✓ In the case of the extensive variable being the Gibbs free energy,

$$\bar{G}_i = \left(\frac{\partial G'}{\partial n_i} \right)_{T, P, n_j, n_k, \dots}$$

8.3.1.3 Partial Molar Quantities

$$\bar{G}_i = \left(\frac{\partial G'}{\partial n_i} \right)_{T, P, n_j, n_k, \dots} = \mu_i$$

- ✓ That is, the **partial molar** Gibbs free energy of a component in a solution **equals** the **chemical potential** of the component in the solution.
- ✓ The relationships among the various state variables developed in the preceding chapters are applicable to the partial molar properties of the components of a system.

8.3.1.3 Partial Molar Quantities

The fundamental equation at constant T and composition gives

$$\left(\frac{\partial G'}{\partial P} \right)_{T, \text{ comp}} = V'$$

where G' is the Gibbs free energy of the system and V' is the volume of the system.

For a variation in n_i , the number of moles of component i in the system, at constant T , P , and n_j ,

$$\left[\frac{\partial}{\partial n_i} \left(\frac{\partial G'}{\partial P} \right)_{T, \text{ comp}} \right]_{T, P, n_j} = \left(\frac{\partial V'}{\partial n_i} \right)_{T, P, n_j}$$

8.3.1.3 Partial Molar Quantities

$$\bar{Q}_i = \left(\frac{\partial Q'}{\partial n_i} \right)_{T, P, n_j, n_k, \dots} \quad \longrightarrow \quad \left(\frac{\partial V'}{\partial n_i} \right)_{T, P, n_j} = \bar{V}_i$$

Since the Gibbs free energy is a thermodynamic state variable, in which case the order of partial differentiation has no influence on the result,

$$\left[\frac{\partial}{\partial n_i} \left(\frac{\partial G'}{\partial P} \right)_{T, \text{comp}} \right]_{T, P, n_j} = \left[\frac{\partial}{\partial P} \left(\frac{\partial G'}{\partial n_i} \right)_{T, P, n_j} \right]_{T, \text{comp}} = \left[\frac{\partial \bar{G}_i}{\partial P} \right]_{T, \text{comp}}$$
$$\quad \quad \quad \downarrow$$
$$\left[\frac{\partial \bar{G}_i}{\partial P} \right]_{T, \text{comp}} = \bar{V}_i$$

8.3.1.3 Partial Molar Quantities

$$\left[\frac{\partial \bar{G}_i}{\partial P} \right]_{T, \text{ comp}} = \bar{V}_i$$

Thus, for the ideal gas A in a mixture of ideal gases,

$$d\bar{G}_A = \bar{V}_A dP$$

The partial molar volume, $\bar{V}_A$, in a gas mixture is

$$\bar{V}_A = \frac{V'}{\sum n_i} = \left(\frac{n_A}{\sum n_i} \right) \left(\frac{RT}{p_A} \right) = X_A \frac{RT}{p_A}$$

8.3.1.3 Partial Molar Quantities

$$p_A = X_A P$$

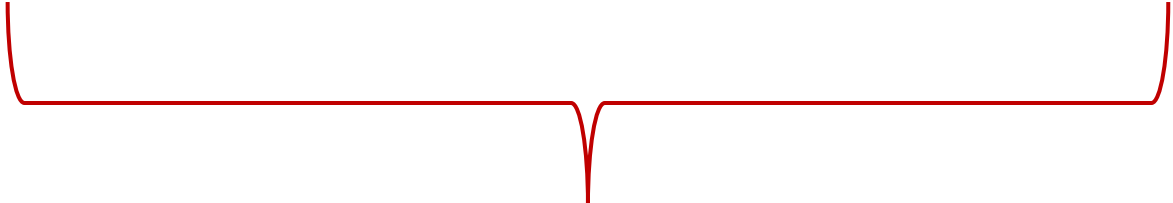
At constant T and composition



$$d\bar{G}_A = \bar{V}_A dP$$

$$dp_A = X_A dP$$

$$\bar{V}_A = X_A \frac{RT}{p_A}$$


$$d\bar{G}_A = \bar{V}_A dP = X_A \frac{RT}{p_A} dP = \frac{RT}{p_A} dp_A$$

8.3.1.3 Partial Molar Quantities

$$d\bar{G}_A = \bar{V}_A dP = X_A \frac{RT}{p_A} dP = \frac{RT}{p_A} dp_A$$

Integration from $p_A=1$ to $p_A=P$ gives

$$p_A = X_A P$$

$$\bar{G}_A = G_A^0 + RT \ln p_A = G_A^0 + RT \ln X_A + RT \ln P$$



$$dG = \frac{RT}{P} dP = RT \ln P$$

Integrating from the standard state $p_A=P_A=1$, $X_A=1$, T to the state p_A , X_A , T for the above equation.

8.3.2 The Enthalpy of Mixing of Ideal Gases

- ✓ For each component gas in a mixture of ideal gases:

$$\bar{G}_A = G_A^0 + RT \ln p_A = G_A^0 + RT \ln X_A + RT \ln P$$

where P is the total pressure of the gas mixture at the temperature T .

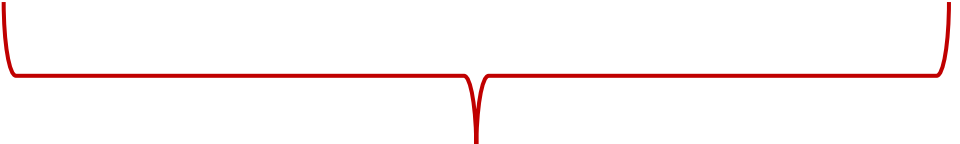
Dividing by T and differentiating with respect to T at constant pressure and composition gives

$$\left(\partial \left(\frac{\bar{G}_i}{T} \right) / \partial T \right)_{P, n_j} = \left(\partial \left(\frac{\bar{G}_i^0}{T} \right) / \partial T \right)_{P, n_j}$$

From the Gibbs-Helmholz equations:

$$\left(\partial \left(\frac{\bar{G}_i^0}{T} \right) / \partial T \right)_{P, n_j} = -\frac{\bar{H}_i^0}{T^2} \quad \left(\partial \left(\frac{\bar{G}_i}{T} \right) / \partial T \right)_{P, n_j} = -\frac{\bar{H}_i}{T^2}$$

8.3.2 The Enthalpy of Mixing of Ideal Gases

$$\left(\partial \left(\frac{\bar{G}_i^0}{T} \right) / \partial T \right)_{P, n_j} = -\frac{\bar{H}_i^0}{T^2} \qquad \left(\partial \left(\frac{\bar{G}_i}{T} \right) / \partial T \right)_{P, n_j} = -\frac{\bar{H}_i}{T^2}$$

$$\bar{H}_i^0 = \bar{H}_i$$

That is, the partial molar enthalpy of ideal gas i in a mixture of ideal gases **equals** the molar enthalpy of pure i .

8.3.2 The Enthalpy of Mixing of Ideal Gases

- ✓ The enthalpy of the gas mixture therefore equals the sum of the enthalpies of the component gases before mixing,

$$\bar{H}_i^0 = \bar{H}_i \quad \Rightarrow \quad \Delta H'^{mix} = \sum_i n_i \bar{H}_i - \sum_i n_i \bar{H}_i^0 = 0$$

$\Delta H'^{mix}$ is the change in the enthalpy caused by the process of mixing.

- ✓ That is, the enthalpy change of mixing (the heat of mixing) of an ideal gas is zero. The zero heat of mixing of ideal gases is a consequence of the fact that the particles of an ideal gas do not interact with one another.

8.3.2 The Enthalpy of Mixing of Ideal Gases

$$\left(\partial \left(\frac{\bar{G}_i}{T} \right) / \partial T \right)_{P, n_j} = \left(\partial \left(\frac{\bar{G}_i^0}{T} \right) / \partial T \right)_{P, n_j}$$

$$\left(\partial \left(\frac{\bar{G}_i^0}{T} \right) / \partial T \right)_{P, n_j} = -\frac{\bar{H}_i^0}{T^2} \quad \left(\partial \left(\frac{\bar{G}_i}{T} \right) / \partial T \right)_{P, n_j} = -\frac{\bar{H}_i}{T^2}$$

Since G_i^0 is, by definition, a function only of temperature, one can see that $\bar{H}_i$ is a function only of temperature. Thus, in addition to being independent of composition, the partial molar enthalpy of an ideal gas, $\bar{H}_i$, is independent of pressure.

8.3.3 The Gibbs Free energy of Mixing of Ideal Gases

- ✓ For each component gas i in a mixture of ideal gases,

$$\bar{G}_i = G^0 + RT \ln p_i$$

For each component gas before mixing,

$$G_i = G^0 + RT \ln P_i$$

p_i is the partial pressure of i in the gas mixture

P_i is the pressure of the pure gas i before mixing

8.3.3 The Gibbs Free energy of Mixing of Ideal Gases

- ✓ The mixing process, being a change of state, can be written as
unmixed components (state 1) → mixed components (state 2)

$$\begin{aligned}\Delta G(1 \rightarrow 2) &= G'(\text{mixture}) - G'(\text{unmixed components}) \\ &= \Delta G'^{\text{mix}} \\ &= \sum_i n_i \bar{G}_i - \sum_i n_i G_i \\ &= \sum_i n_i RT \ln \left(\frac{p_i}{P_i} \right)\end{aligned}$$

The value of $\Delta G'^{\text{mix}}$ depends on the value of p_i and P_i for each gas.

8.3.3 The Gibbs Free energy of Mixing of Ideal Gases

- ✓ If, before mixing, the gases are all at the same pressure(i.e., if $P_i = P_j = P_k = \dots$) and mixing is carried out at total constant volume such that the total pressure of the mixture, P_{mix} , equals the initial pressures of the gases before mixing, then, since $p/P = X_i$,

$$\Delta G'^{mix} = \sum_i n_i RT \ln X_i$$

- ✓ Since the values of X_i are less than unity, $\Delta G'^{mix}$ is a negative quantity, which corresponds with the fact that the mixing of ideal gases is a spontaneous process.

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8.4 Deviation from Ideality and Equations of State for Real Gases

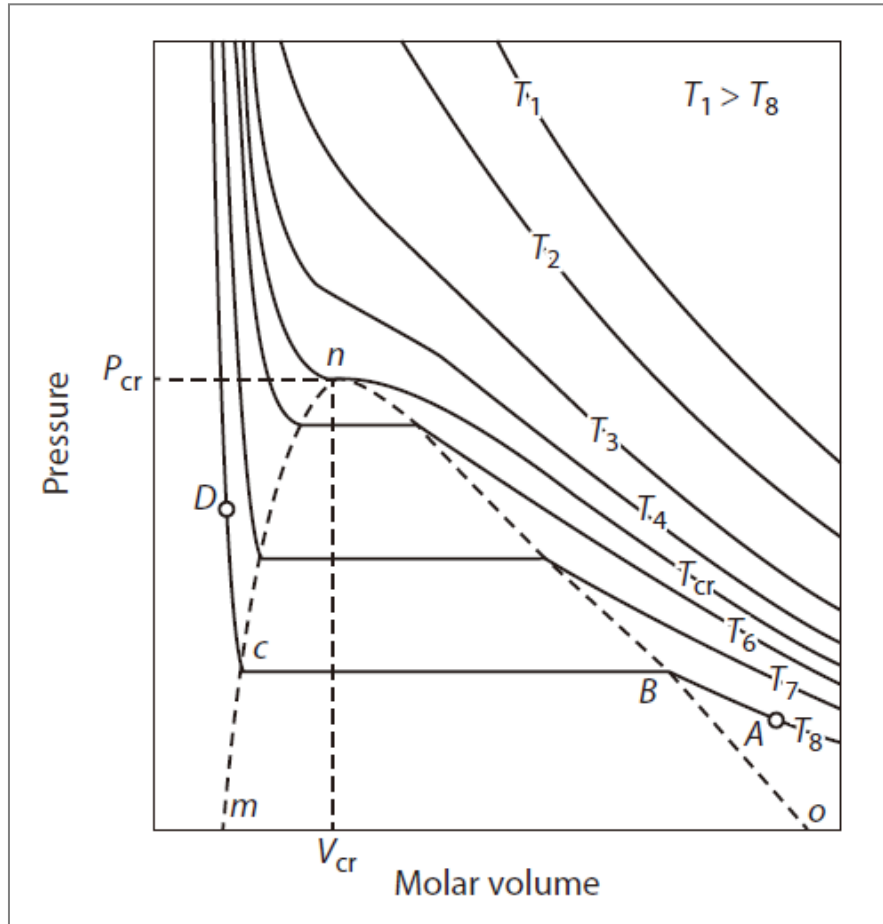


Figure 8.1 P - V isotherms for a typical real gas.

As the temperature of the gas is decreased from the high temperature T_1 , the shape of the P - V isotherms changes, and, eventually, a critical value of $T = T_{cr}$ is reached at which, at some fixed critical pressure, P_{cr} , and fixed molar volume, V_{cr} , a horizontal inflection occurs on the isotherm; that is,

$$\left(\frac{\partial P}{\partial V} \right)_{T_{Cr}} = 0 \quad \& \quad \left(\frac{\partial^2 P}{\partial V^2} \right)_{T_{Cr}} = 0$$

8.4 Deviation from Ideality and Equations of State for Real Gases

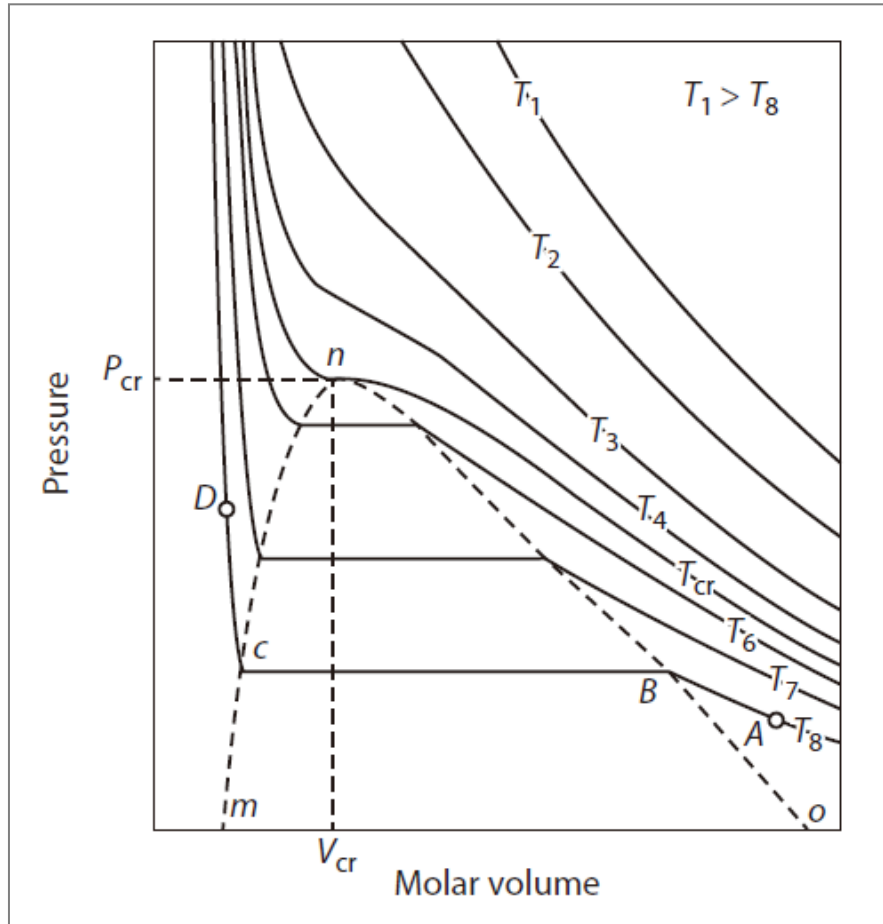


Figure 8.1 P - V isotherms for a typical real gas.

At temperatures less than T_{cr} two phases exist in equilibrium: a vapor phase and a liquid phase.

For example, if 1 mole of vapor, initially in the state A is isothermally compressed at T_8 , the state of the vapor moves along the isotherm toward the state B .

8.4 Deviation from Ideality and Equations of State for Real Gases

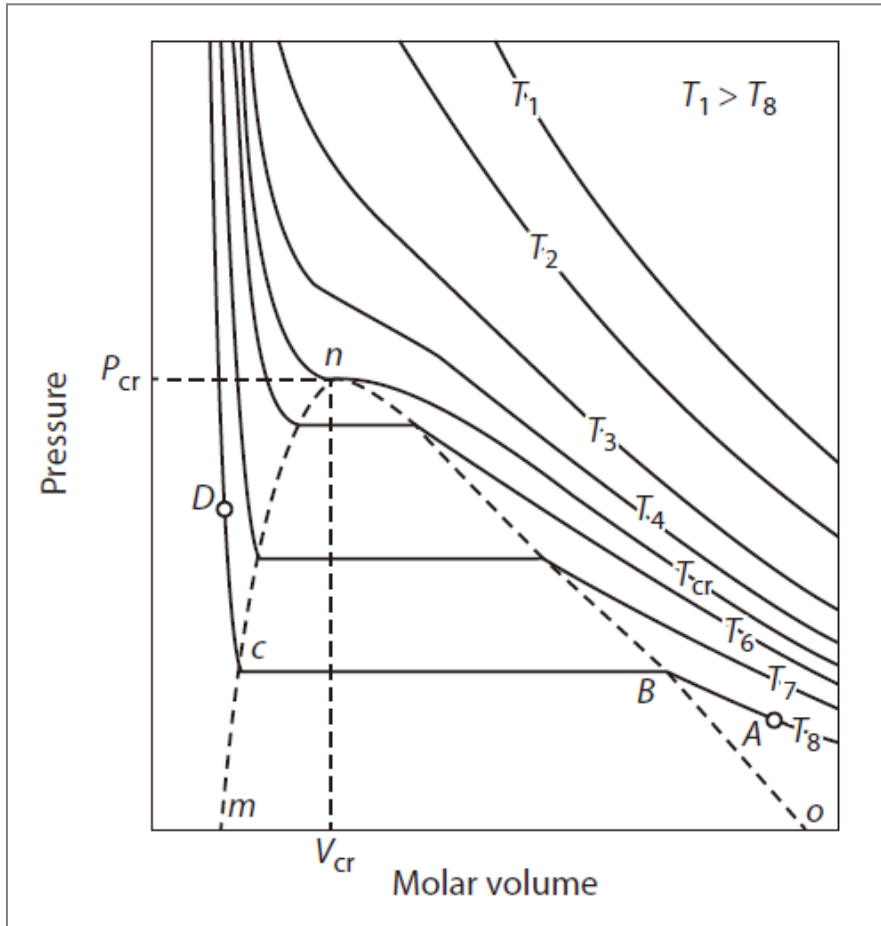
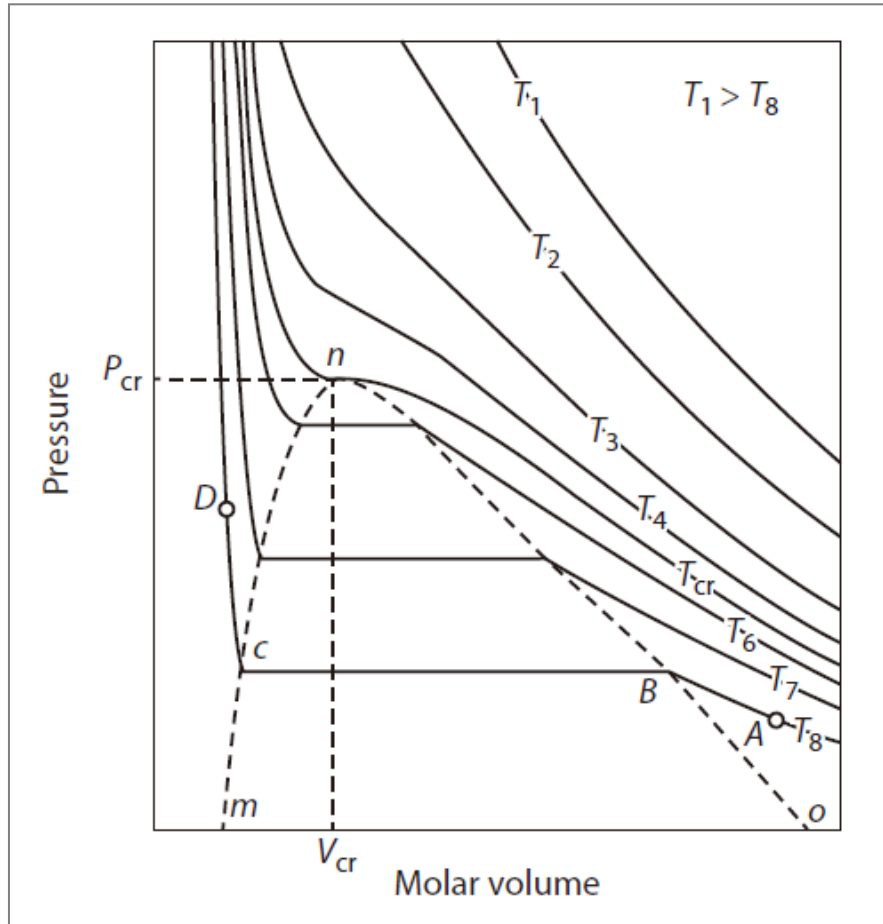


Figure 8.1 P - V isotherms for a typical real gas.

At **B**, the pressure of the vapor is the saturated vapor pressure of the liquid at T_8 , and further compression of the system causes condensation of the vapor and the consequent appearance of the liquid phase.

The liquid phase, which is in equilibrium with the vapor, first appears when the vapor reaches **B** on the T_8 isotherm. V_C is the molar volume of the liquid at P_C and T_8 .

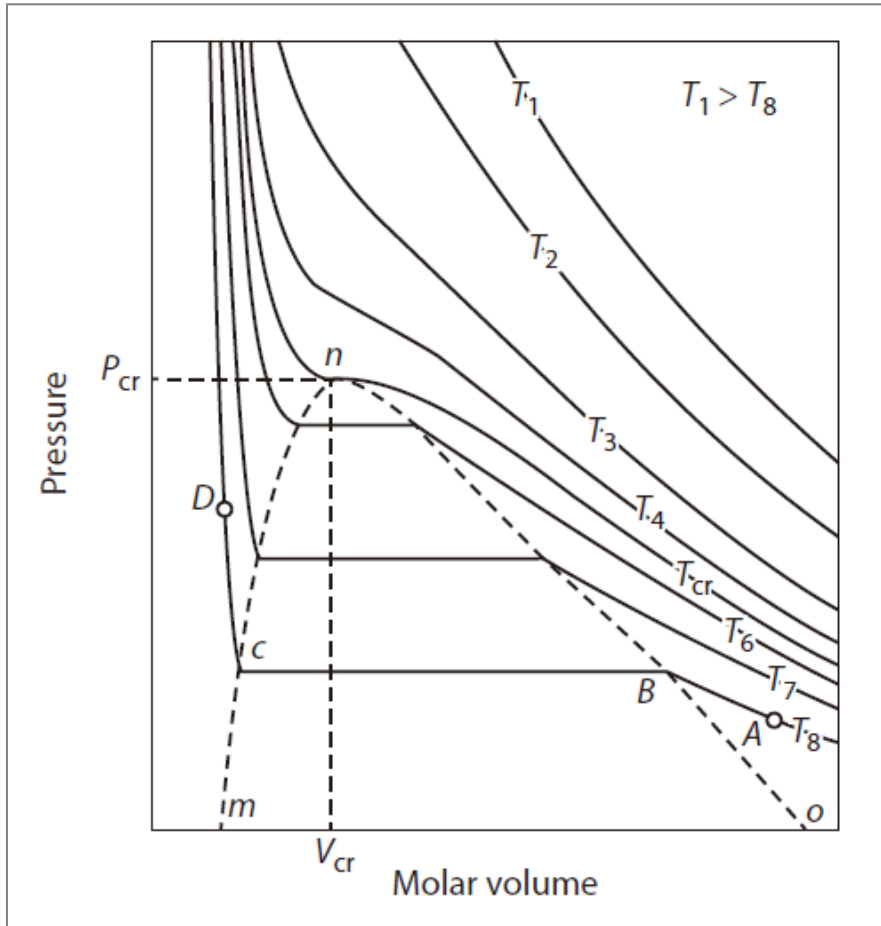
8.4 Deviation from Ideality and Equations of State for Real Gases



Further compression causes further condensation, during which the states of the liquid and vapor phases remain fixed at C and B , respectively, and the total volume of the system, which is determined by the relative proportions of the liquid and vapor phases, moves along the horizontal line from B to C .

Figure 8.1 P - V isotherms for a typical real gas.

8.4 Deviation from Ideality and Equations of State for Real Gases



Eventually, condensation is complete, and the system exists as 100% liquid in the state C. Further increase in pressure moves the state of the system along the isotherm toward the state D.

Figure 8.1 P - V isotherms for a typical real gas.

8.4 Deviation from Ideality and Equations of State for Real Gases

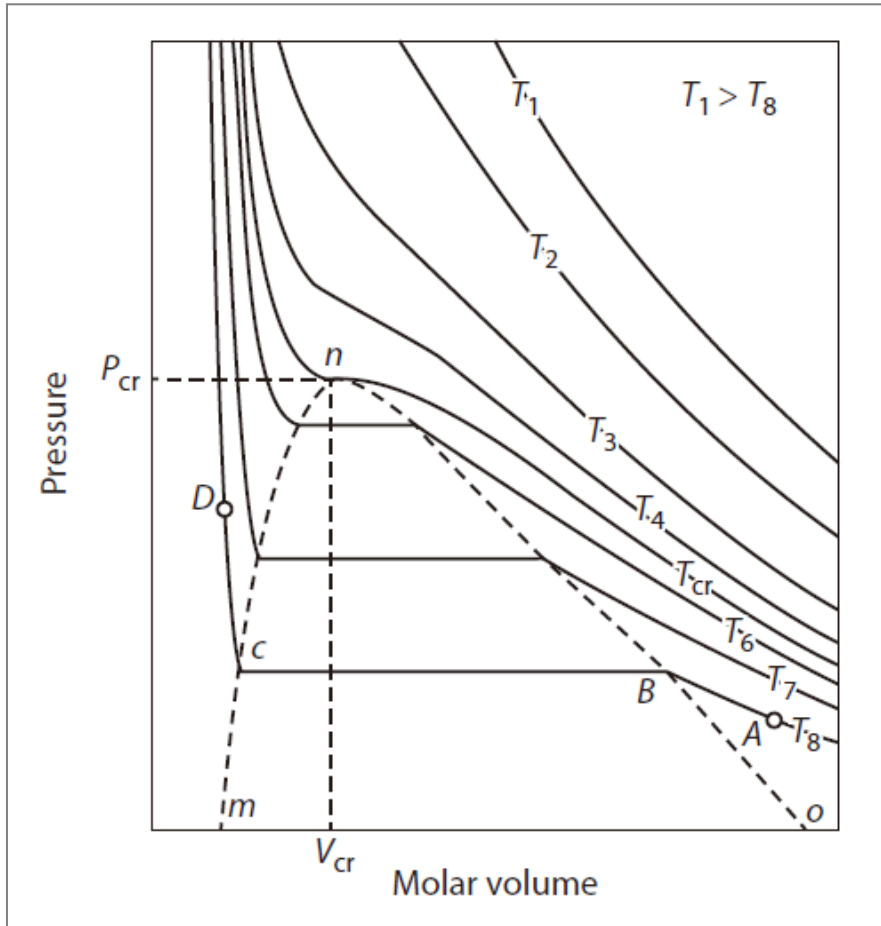


Figure 8.1 P - V isotherms for a typical real gas.

The large value of $-(\partial P/\partial V)_T$ in the range of liquid states and the smaller value of $-(\partial P/\partial V)_T$ in the range of vapor states indicate the **low compressibility** of the liquid phase and the **high compressibility** of the vapor phase.

The isothermal compressibility equals

$$\beta_T = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T$$

8.4 Deviation from Ideality and Equations of State for Real Gases

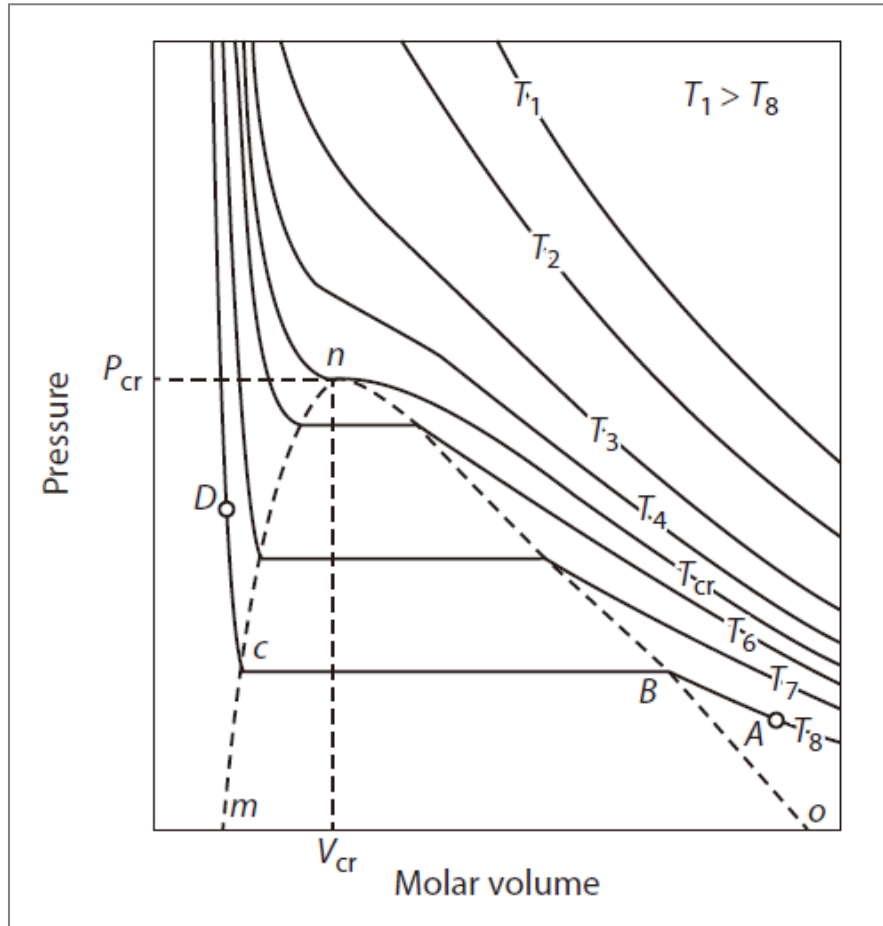


Figure 8.1 P - V isotherms for a typical real gas.

As the temperature is increased from T_8 to T_{cr} , the molar volume of the liquid in equilibrium with the vapor (corresponding to the point C at T_8) progressively increases and the molar volume of the vapor in equilibrium with the liquid (corresponding to the point B at T_8) progressively decreases. Thus, as the temperature is increased toward T_{cr} , the vapor in equilibrium with liquid becomes more dense, and the liquid in equilibrium with the vapor becomes less dense.

8.4 Deviation from Ideality and Equations of State for Real Gases

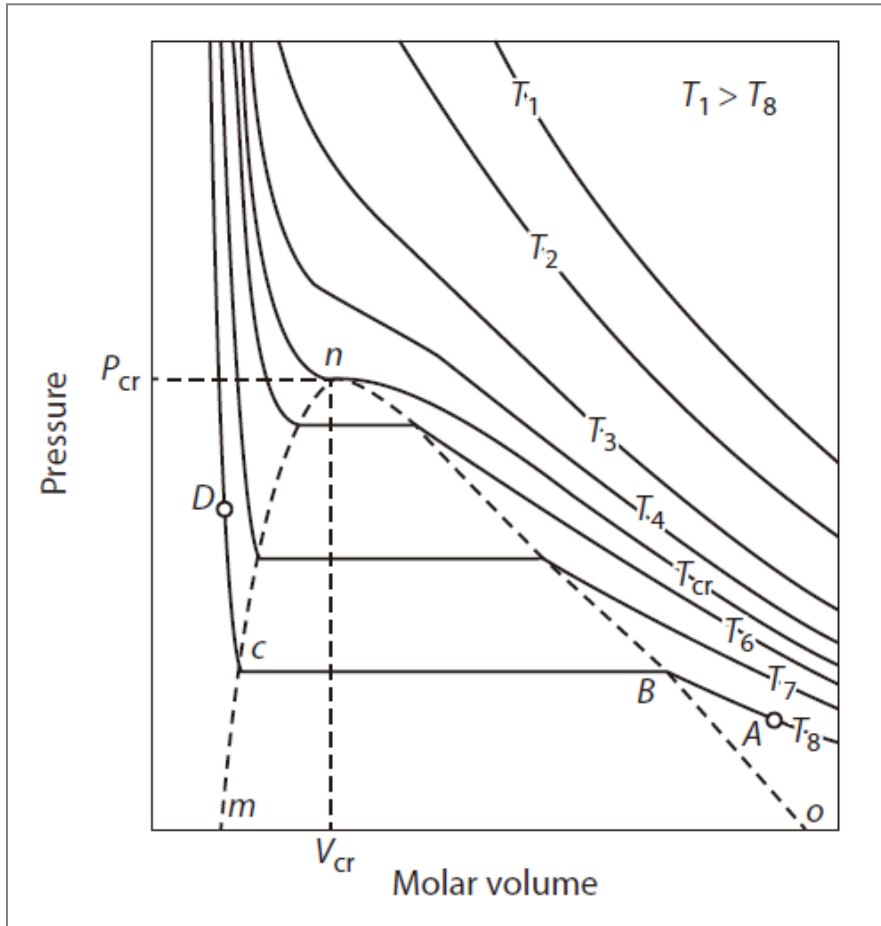
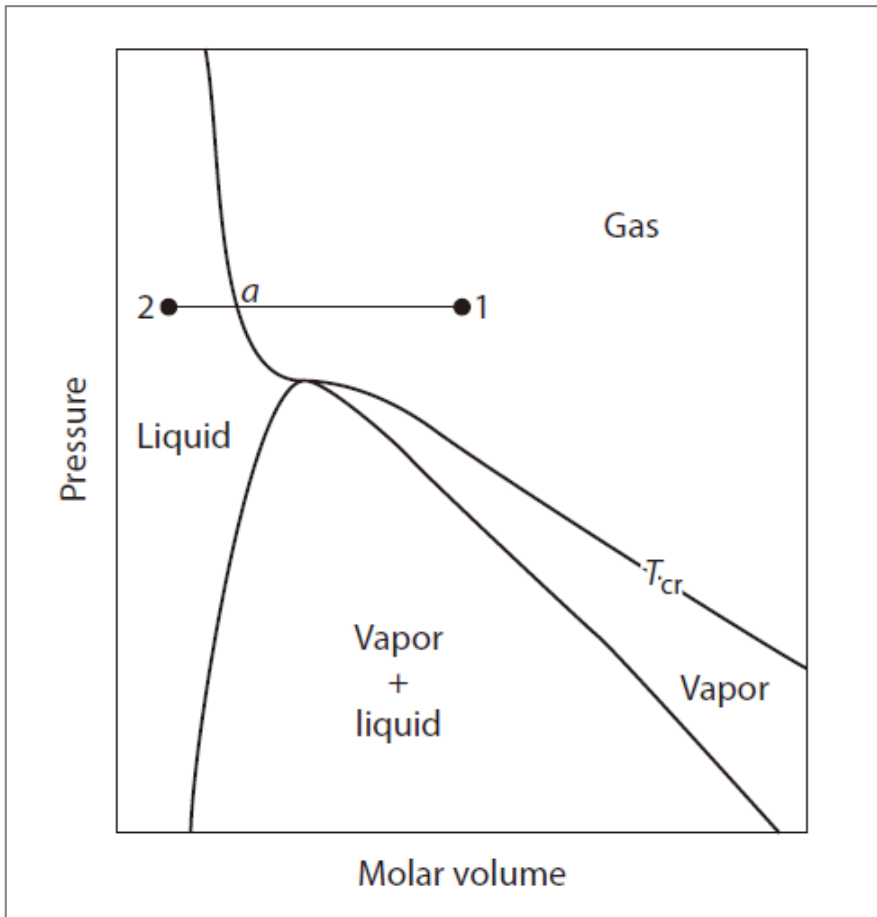


Figure 8.1 P - V isotherms for a typical real gas.

Eventually, when T_{cr} is reached, the molar volumes of the coexisting phases coincide at the state P_{cr}, T_{cr} . Thus, the critical point coincides with the equilibrium state in which the molar volume of the liquid **equals** the molar volume of the vapor.

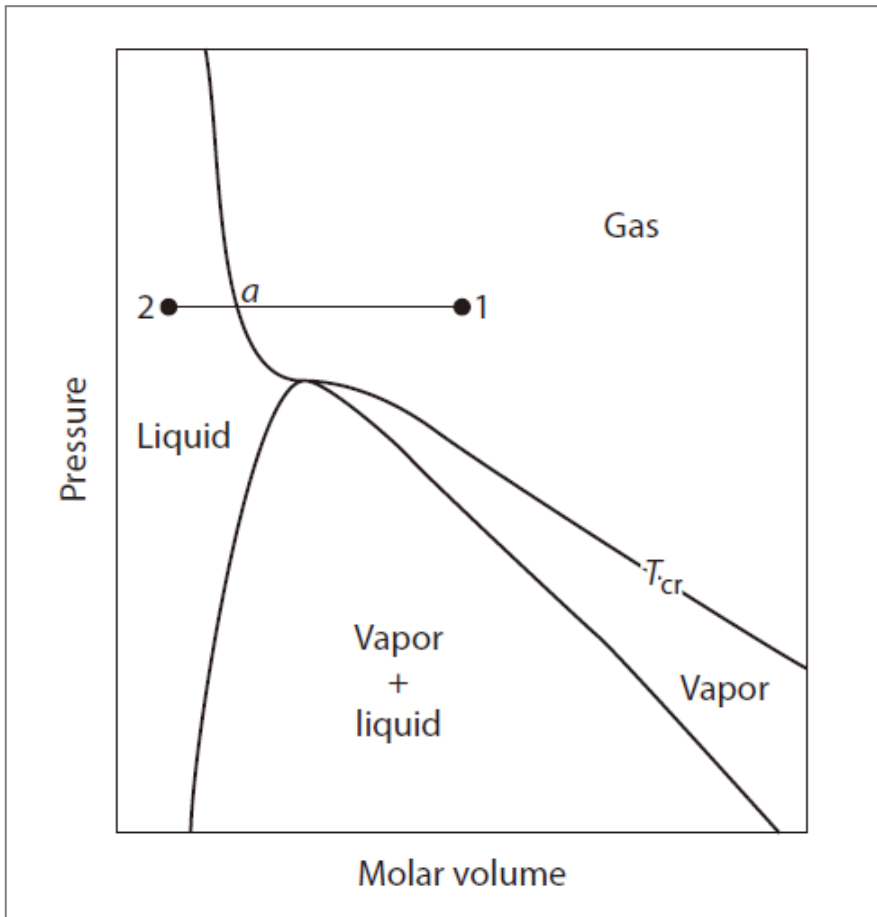
8.4 Deviation from Ideality and Equations of State for Real Gases



At temperatures greater than T_{cr} , distinct two-phase equilibrium (involving two phases separated by a boundary, across which the properties of the system change abruptly) does not occur, and thus, the gaseous state cannot be liquefied by isothermal compression at temperatures greater than T_{cr} .

Figure 8.2 schematic P - V diagram showing the fields of phase stability of a typical real gas.

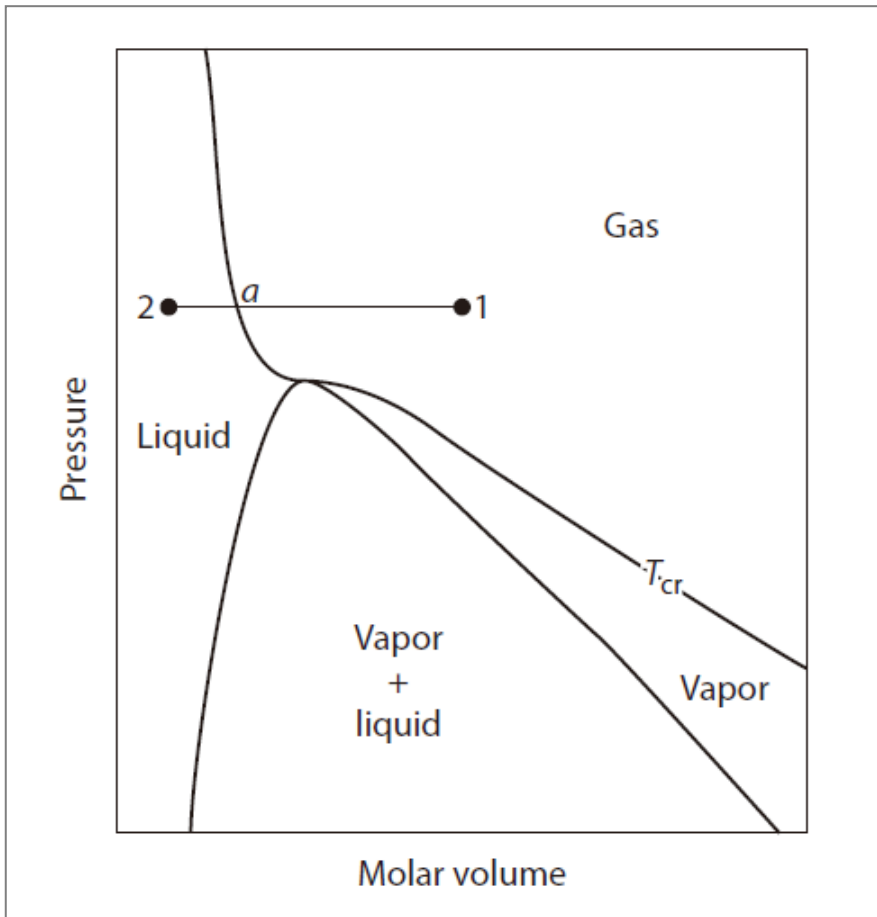
8.4 Deviation from Ideality and Equations of State for Real Gases



Since the vapor can be condensed by isothermal compression at temperatures lower than T_{cr} , the **critical isotherm** provides a distinction between the **gaseous** and **vapor** states and defines the gaseous state phase field.

Figure 8.2 schematic P - V diagram showing the fields of phase stability of a typical real gas.

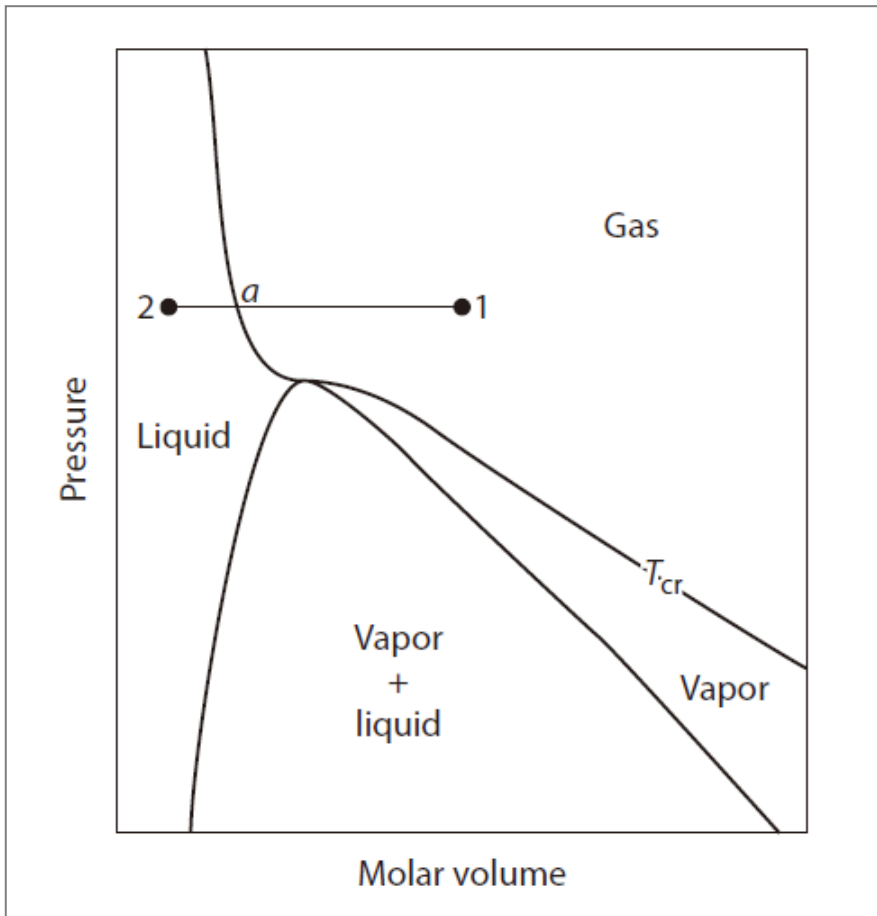
8.4 Deviation from Ideality and Equations of State for Real Gases



Note, however, that along the critical isotherm, **there is no physical distinction** between the vapor and gaseous phase. Liquefaction of a gas requires that the gas be cooled. Consider the isobaric process path $1 \rightarrow 2$. According to this path, which represents the cooling of the gas at constant pressure, the phase change gas $\rightarrow$ liquid occurs at the point a , at which the temperature falls below T_{cr} .

Figure 8.2 schematic P - V diagram showing the fields of phase stability of a typical real gas.

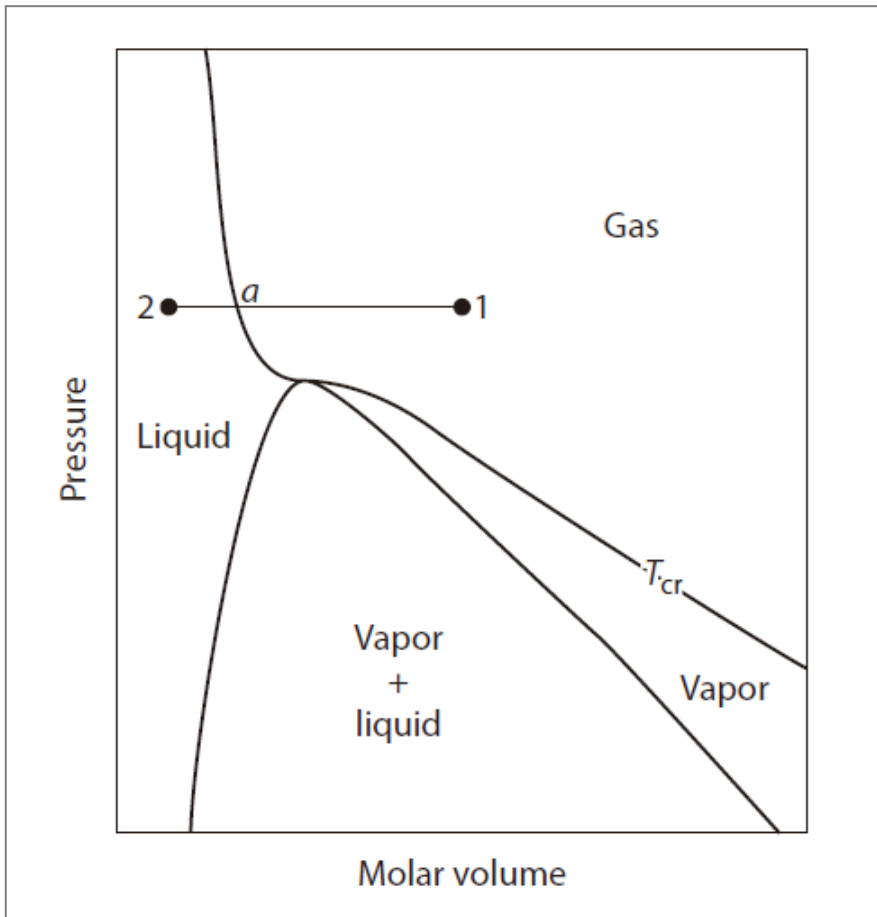
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In fact, at pressures greater than P_{cr} , the critical-temperature isotherm has no physical significance. In passing from the state 1 to the state 2, the molar volume of the system progressively decreases, and hence, the density of the system progressively increases.

Figure 8.2 schematic P - V diagram showing the fields of phase stability of a typical real gas.

8.4 Deviation from Ideality and Equations of State for Real Gases



No phase separation occurs between the states 1 and 2, and the system in state 2 can equivalently be regarded as being a liquid of normal density or a gas of high density and in state 1 can be regarded as being a gas of normal density or a liquid of low density.

Figure 8.2 schematic P - V diagram showing the fields of phase stability of a typical real gas.

8.4 Deviation from Ideality and Equations of State for Real Gases

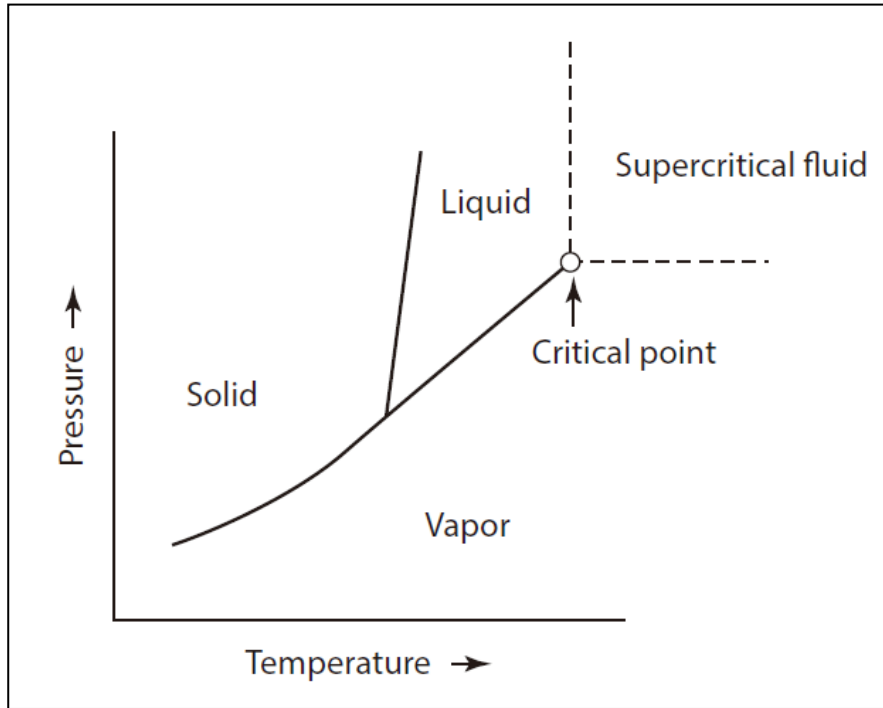


Figure 8.3 the variations, with pressure, of the compressibility factors of several gases at 0° C.

Physically, no distinction can be made between the liquid and gaseous states at pressures greater than P_{cr} and consequently, the system existing in these states is called a supercritical fluid.

In the P - T phase diagram, the critical point exists at the termination of the liquid–vapor coexistence equilibrium curve, where $P=P_{cr}$ and $T=T_{cr}$.

8.4 Deviation from Ideality and Equations of State for Real Gases

- ✓ The **deviation** of a real gas from ideal behavior can be measured as the deviation of the **compressibility factor** from unity. The compressibility factor, **Z**, is defined as

$$Z = \frac{PV}{RT}$$

which has the value of 1 for a perfect gas in all states of existence

- ✓ **Z** itself is a function of state of the system and, thus, is dependent on any two chosen dependent variables; for example, **Z** = **Z(P,T)**.

8.4 Deviation from Ideality and Equations of State for Real Gases

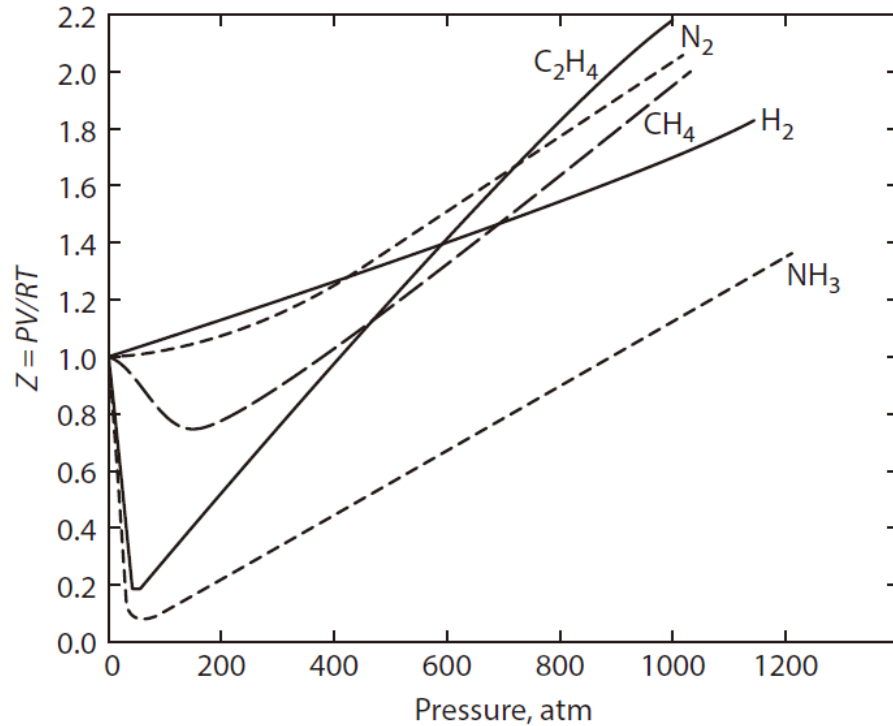


Figure 8.4 The variations, with pressure, of the compressibility factors of several gases at 0°C.

If it is replotted as Z versus the reduced pressure, P_R (where $P_R = P/P_{cr}$), for fixed values of the reduced temperature, T_R ($=T/T_{cr}$), it is found that all gases lie on a single curve.

8.4 Deviation from Ideality and Equations of State for Real Gases

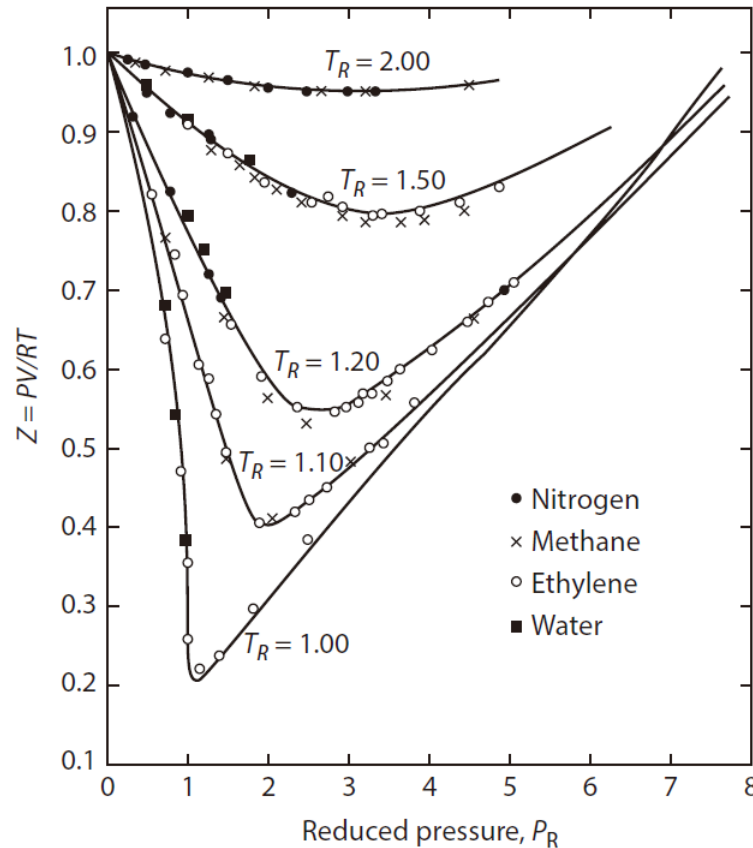


Figure 8.5 The variations of the compressibility factors of several gases with reduced pressure at several reduced temperatures, displaying the law of corresponding states.

All gases obey the same equation of state when expressed in terms of the reduced variables P_R , T_R , and V_R instead of P , T , and V .

Outlines

8.1 Introduction

8.2 The P - V - T Relationship of Gases

8.3 The Thermodynamic Properties of Ideal Gases and Mixtures of Ideal Gases

- 8.3.1 Mixtures of Ideal Gases
- 8.3.2 The Enthalpy of Mixing of Ideal Gases
- 8.3.3 The Gibbs Free Energy of Mixing of Ideal Gases
- 8.3.4 The Entropy of Mixing of Ideal Gases

8.4 Deviation from Ideality and Equations of State for Real Gases

8.5 The Van Der Waals Fluid

8.6 Other Equations of State for Nonideal Gases

8.7 Further Thermodynamic Treatment of Nonideal Gases

8.8 Summary

8.5 The Van Der Waals Fluid

- ✓ Recall that an ideal gas obeys the ideal gas law and has an internal energy, U , which is a function only of temperature. An ideal gas is considered to consist of volumeless particles that do not interact with one another, the energy of which is entirely the translational energy of motion of the constituent particles.
- ✓ Attempts to derive equations of state for real gases have modified the ideal gas equation by taking into account two considerations:
 - The particles of a real gas occupy a finite volume.
 - The particles of a real gas interact with one another.

8.5 The Van Der Waals Fluid

- The particles of a real gas occupy a finite volume.
- The particles of a real gas interact with one another.

The magnitude of the importance of these two considerations depends on the state of the gas.

- If the molar volume of the gas is large, then the volume fraction occupied by the particles themselves is small, and the magnitude of this effect on the behavior of the gas will be correspondingly small.
- Similarly, as the molar volume increases, the average distance between the particle increases, and thus, the effect of interactions between particles on the behavior of the gas decreases.

8.5 The Van Der Waals Fluid

- ✓ For a fixed number of moles of gas, an increase in the molar volume corresponds to a decrease in the **density**, n/V' . These states of existence occur at low pressure and high temperature, as can be seen from the ideal gas equation; that is,

$$\frac{n}{V'} = \frac{P}{RT}$$

- ✓ Thus, approach toward ideal behavior is to be expected as the pressure is decreased and the temperature is increased.

8.5 The Van Der Waals Fluid

- ✓ The most celebrated equation of state for nonideal gases, which was derived from considerations 1 and 2, is the van der Waals equation (Johannes Diderik van der Waals, 1837–1923), which, for 1 mole of gas, is written as

$$\left(P + \frac{a}{V^2} \right) (V - b) = RT$$

P is the measured pressure of the gas

a/V^2 is a correction term for the interactions which occur among the particles of the gas

V is the measured volume of the gas

b is a correction term for the finite volume of the particles

8.5 The Van Der Waals Fluid

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

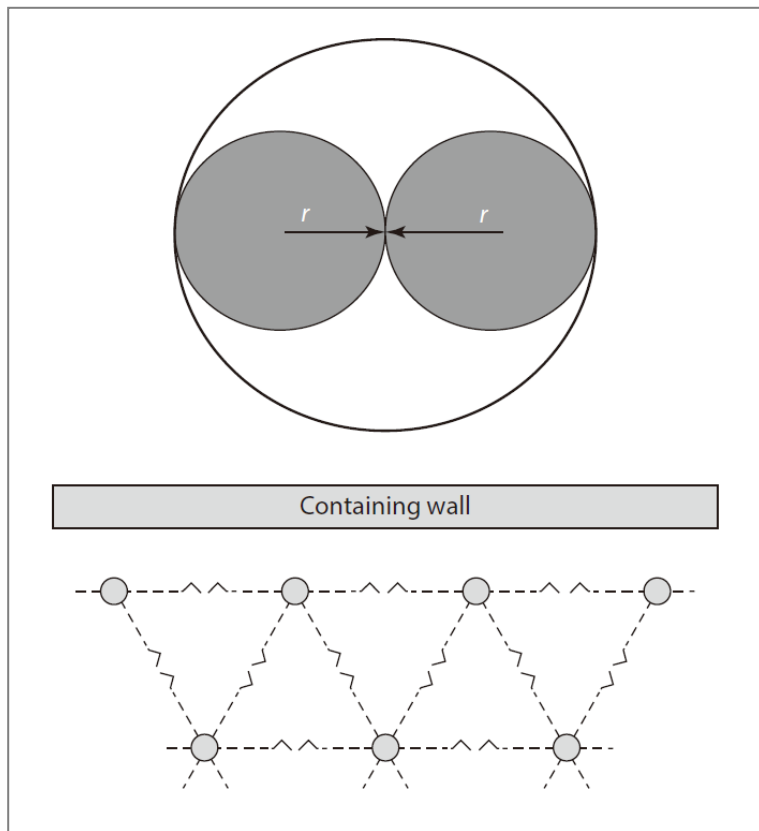
The term b is determined by considering a collision between two spherical particles. Two particles, of radius r , collide when the distance between their centers decreases to a value less than $2r$, and at the point of collision, the particles exclude a volume of to all the other particles.

The volume excluded per particle is:

$$\frac{4}{3}\pi(2r)^3$$

8.5 The Van Der Waals Fluid

$$\frac{1}{2} \times \frac{4}{3} \pi (2r)^3 = 4 \times \frac{4}{3} \pi r^3 = 4 \times \text{the volume of one particle}$$



$$\frac{4}{3} \pi (2r)^3$$

Figure 8.6 (a) Illustration of the **volume excluded** when two spherical atoms contact; (b) the **interactions** among atoms in a gas phase, showing that the interactions cause a lower pressure to be exerted on the walls of the container.

8.5 The Van Der Waals Fluid

$$\frac{1}{2} \times \frac{4}{3} \pi (2r)^3 = 4 \times \frac{4}{3} \pi r^3 = 4 \times \text{the volume of one particle}$$

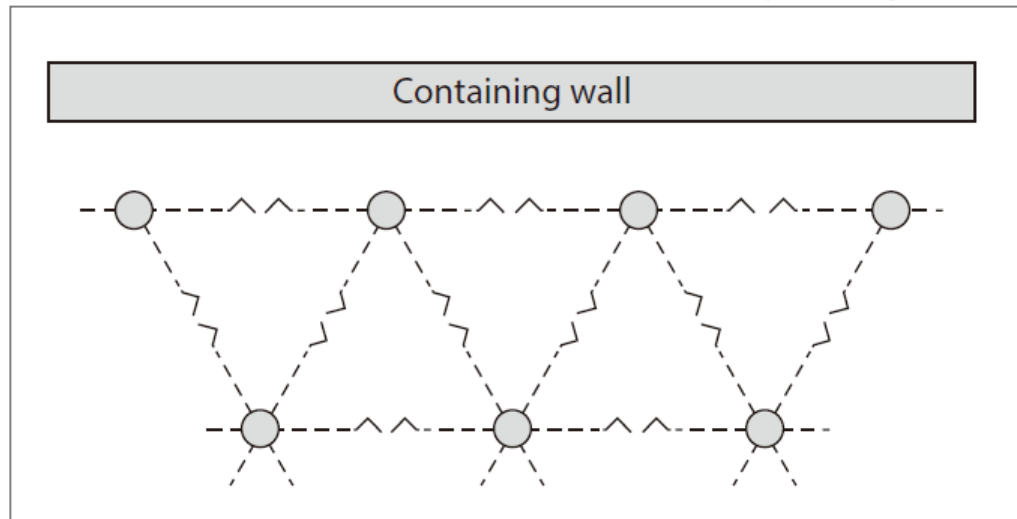
- ✓ The **volume excluded** is thus four times the volume of all of the particles present and **has the value b** .
- ✓ In 1 mole of gas, the volume $(V - b)$ is that available for the motion of the particles of the gas and is the molar volume which the gas would have were the gas ideal—that is, if the particles were volumeless.

8.5 The Van Der Waals Fluid

- ✓ The long-range attractive forces operating between the gas particles decrease the pressure exerted on the containing wall to a value less than that which would be exerted in the absence of the forces.
- Van der Waals considered the following: the particles in the “layer” adjacent to the containing wall experience a net inward pull due to interaction with the particles in the next adjacent layer.
- These attractive forces give rise to the phenomenon of internal pressure, and the magnitude of the net inward pull (i.e., the decrease in the pressure exerted by the gas on the containing wall) is proportional to the number of particles in the “surface layer” and to the number of particles in the “next-to-the-surface layer.”

8.5 The Van Der Waals Fluid

- Both of these quantities are proportional to the density of the gas, n/V , and hence, the net inward pull is proportional to the square of the density of the gas, or, for 1 mole of gas, equal to a/V^2 , where a is a constant.
- ✓ Thus, if P is the measured pressure of the gas, $P + a/V^2$ is the pressure which the gas would exert on the containing wall if the gas were ideal—that is, in the absence of interactions among the particles.



8.5 The Van Der Waals Fluid

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT \quad \Rightarrow \quad PV^3 - (Pb + RT)V^2 + aV - ab = 0$$

Since this equation is cubic in V , there are three roots. Plotting V as a function of P for different values of T gives the family of isotherms.

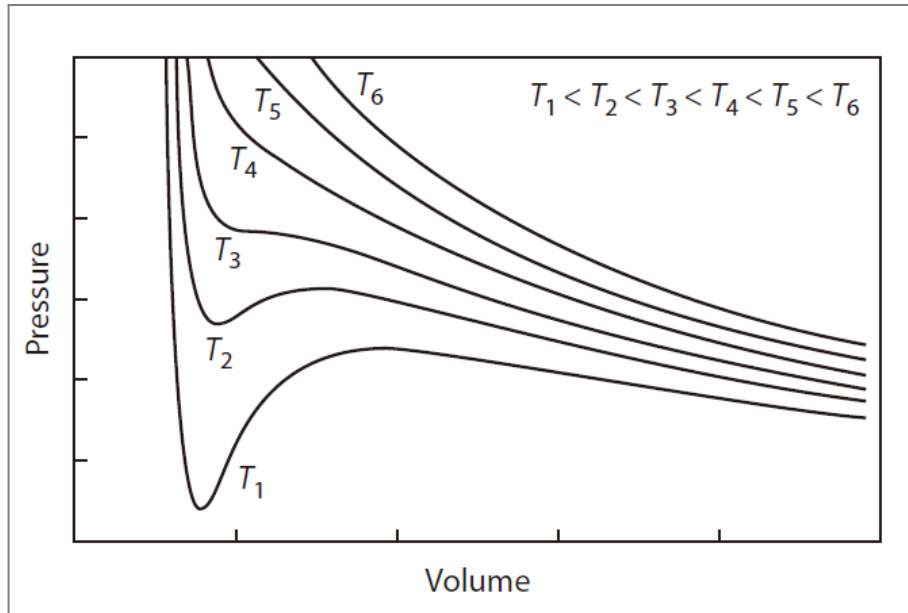


Figure 8.7 The isothermal variation of V with P for a van der Waals fluid at several temperatures.

As the temperature is increased from T_1 , the minimum and the maximum approach one another until, at a temperature designated as T_{cr} , they coincide and produce **a horizontal inflection on the P - V curve.**

8.5 The Van Der Waals Fluid

- ✓ For the critical point, $T = T_{cr}$, $P = P_{cr}$ and $V = V_{cr}$ and the van der Waals equation gives

$$\begin{aligned} P_{cr} &= \frac{RT_{cr}}{(V_{cr} - b)} - \frac{a}{V_{cr}^2} \\ \left(\frac{\partial P}{\partial V} \right)_{T_{cr}} &= -\frac{RT_{cr}}{(V_{cr} - b)^2} + \frac{2a}{V_{cr}^3} = 0 \\ \left(\frac{\partial^2 P}{\partial V^2} \right)_{T_{cr}} &= \frac{2RT_{cr}}{(V_{cr} - b)^3} - \frac{6a}{V_{cr}^4} = 0 \end{aligned} \quad \Rightarrow \quad \begin{aligned} T_{cr} &= \frac{8a}{27bR} \\ V_{cr} &= 3b \\ P_{cr} &= \frac{a}{27b^2} \end{aligned}$$

8.5 The Van Der Waals Fluid

- ✓ Thus, the constants **a** and **b** for any gas can be evaluated from knowledge of the values of T_{cr} and P_{cr} . Alternatively, the values of a and b can be obtained by fitting the van der Waals equation to experimentally measured variations of V with T and P for real gases.

Table 8.1 The Critical States, van der Waals Constants, and Values of Z at the Critical Points for Several Gases

Gas	T_{cr}, K	P_{cr}, atm	$V_{cr}, cm^3/mole$	$a, \frac{l^2 \cdot atm}{mole^2}$	$b, liters/mole$	Z_{cr}
He	5.3	2.26	57.6	0.0341	0.0237	0.299
H ₂	33.3	12.8	65.0	0.2461	0.0267	0.304
N ₂	126.1	33.5	90.0	1.39	0.0391	0.292
CO	134.0	35.0	90.0	1.49	0.0399	0.295
O ₂	153.4	49.7	74.4	1.36	0.0318	0.293
CO ₂	304.2	73.0	95.7	3.59	0.0427	0.280
NH ₃	405.6	111.5	72.4	4.17	0.0371	0.243
H ₂ O	647.2	217.7	45.0	5.46	0.0305	0.184

8.5 The Van Der Waals Fluid

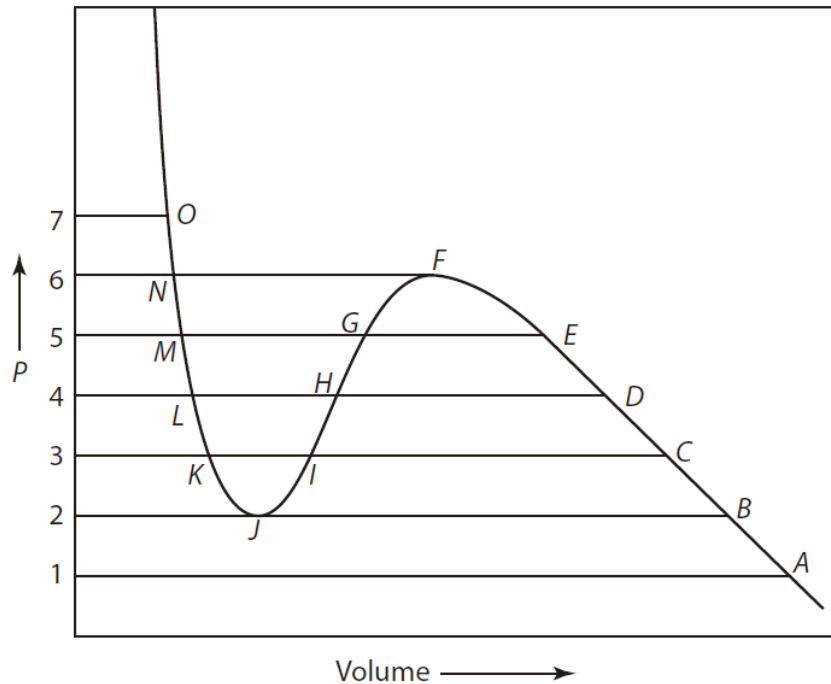


Figure 8.8
the isothermal variation, with pressure,
of the volume of a van der Waals fluid at
a temperature below the critical
temperature.

Any increase in the pressure exerted on a system must cause a decrease in the volume of the system, $(\partial P / \partial V)_T < 0$. This is a condition of intrinsic stability. In Figure 8.8, this condition is violated over the portion **JIHGF**, which means that this portion of the curve is a region of instability.

8.5 The Van Der Waals Fluid

- ✓ The equilibrium states of the system at this temperature can be obtained from a consideration of the variation of the Gibbs free energy with P along the isotherm.

$$dG = -SdT + VdP$$

The variation of G with P at constant T is $dG = VdP$, and integration of this equation between the state (P, T) and (P_A, T) gives

$$G(P, T) - G(P_A, T) = \int_{P_A}^P VdP$$

$$G = G_A + \int_{P_A}^P VdP$$

8.5 The Van Der Waals Fluid

$$G = G_A + \int_{P_A}^P V dP$$

If an arbitrary value is assigned to G_A , then graphical integration of the integral from Figure 8.8 allows the variation of G with P , to be plotted corresponding to the variation of V with P in Figure 8.8.

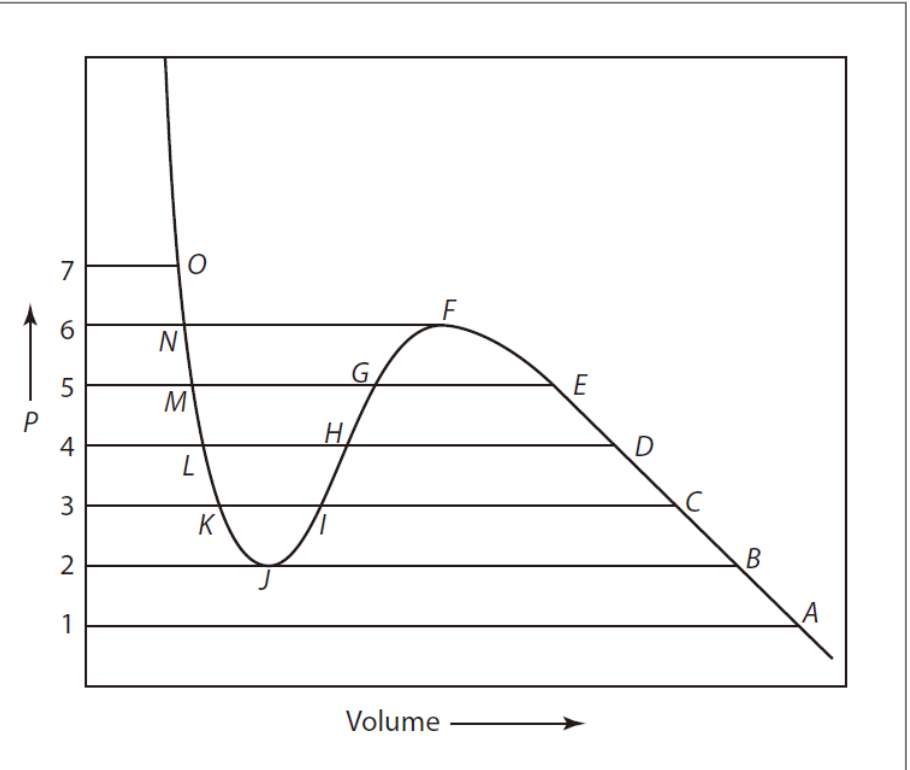
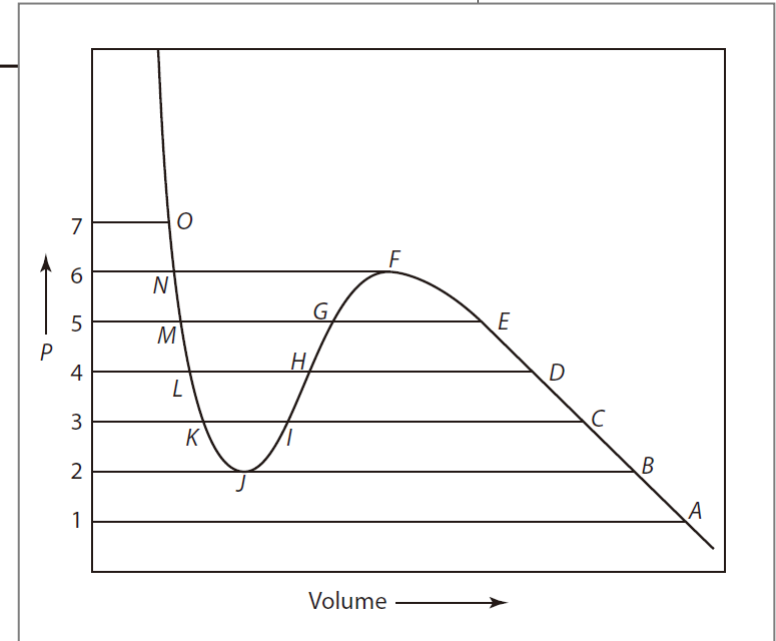


Figure 8.8
the isothermal variation, with pressure,
of the volume of a van der Waals fluid at
a temperature below the critical
temperature.

8.5 The Van Der Waals Fluid

Table 8.2 Graphical Integration of Figure 8.7

$G_B = G_A + \int_{P_A}^{P_B} V dP$	$= G_A + \text{area } 1AB2$
G_C	$= G_A + \text{area } 1AC3$
G_D	$= G_A + \text{area } 1AD4$
G_E	$= G_A + \text{area } 1AE5$
G_F	$= G_A + \text{area } 1AF6$
G_G	$= G_A + \text{area } 1AE5 + \text{area } EFG$
G_H	$= G_A + \text{area } 1AD4 + \text{area } DFH$
G_I	$= G_A + \text{area } 1AC3 + \text{area } CFI$
G_J	$= G_A + \text{area } 1AB2 + \text{area } BFJ$
G_K	$= G_A + \text{area } 1AC3 + \text{area } CFI - \text{area } IJK$
G_L	$= G_A + \text{area } 1AD4 + \text{area } DFH - \text{area } HJL$
G_M	$= G_A + \text{area } 1AE6 + \text{area } EFG - \text{area } GJM$
G_N	$= G_A + \text{area } 1AF6 - \text{area } FJN$
G_O	$= G_A + \text{area } 1AF6 - \text{area } FJN + \text{area } 6NO7$



8.5 The Van Der Waals Fluid

- ✓ As the pressure is increased from P_1 , the value of G increases. At pressures greater than P_2 , **three states** of existence become available to the system. For example, at P_3 , the three states are given by the points **I, K, and C**. The stable, or equilibrium, state is that with the lowest Gibbs free energy, and hence, over the range of pressure from P_2 to P_4 , the stable states lie on the line **BCD**. As the pressure is increased above P_4 , the state with the lowest Gibbs free lies on the curve **LMN**.

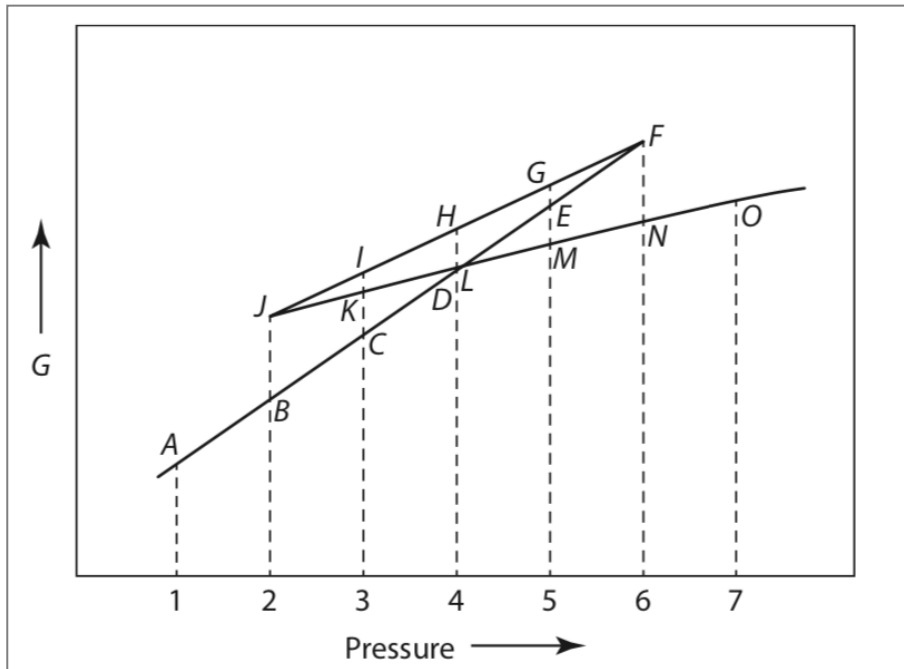


Figure 8.9
schematic representation of the variation, with pressure, of the molar Gibbs free energy of a van der Waals fluid at a constant temperature lower than the critical temperature. region $FGHIJ$ is an instability region. F and J demark the limits of phase stability, sometimes called the spinodals.

8.5 The Van Der Waals Fluid

- ✓ The change of stability at P_4 corresponds to a change of phase at this point; that is, at pressures less than P_4 , one phase is stable (vapor), and at pressures greater than P_4 , another phase is stable (liquid).

At low pressures ($P < P_4$), the system exists as a **vapor**

At high pressures ($P > P_4$), it exists as a **liquid**.

Fig. 8.9

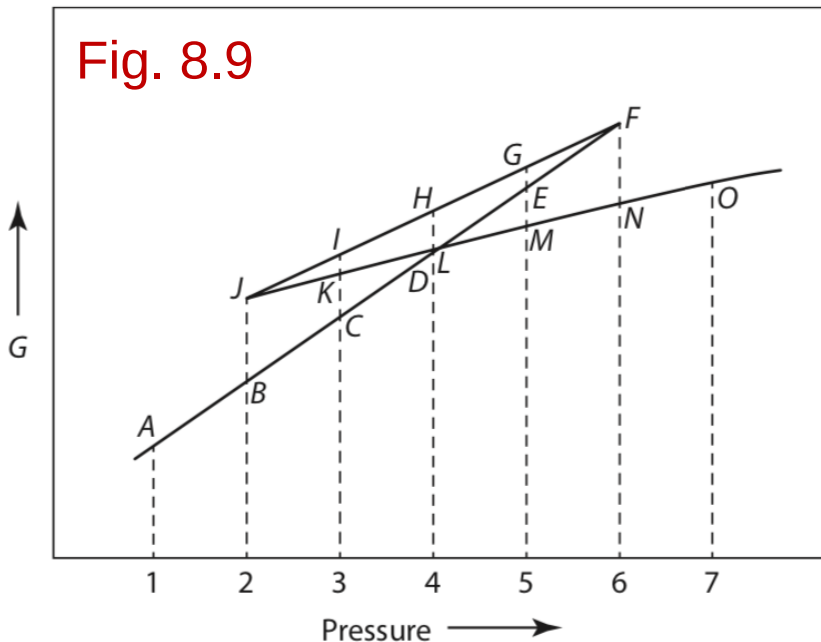
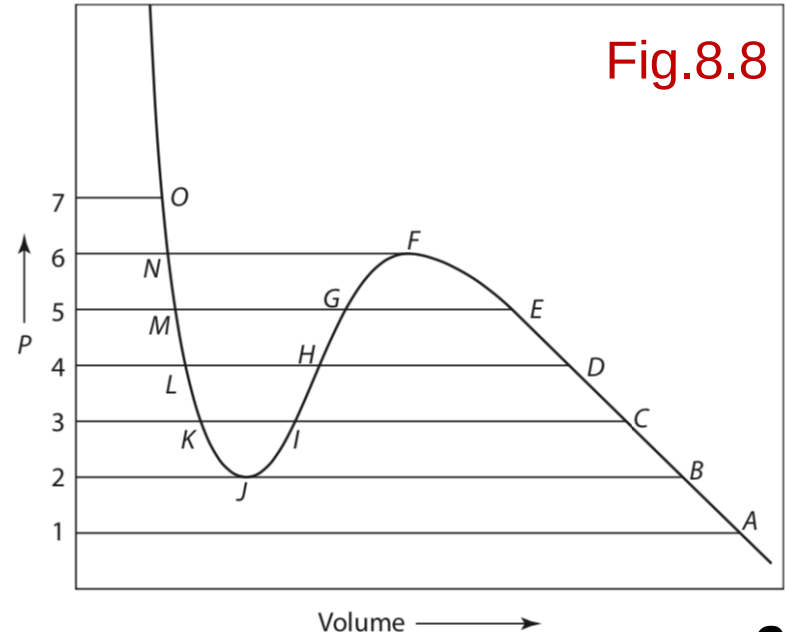
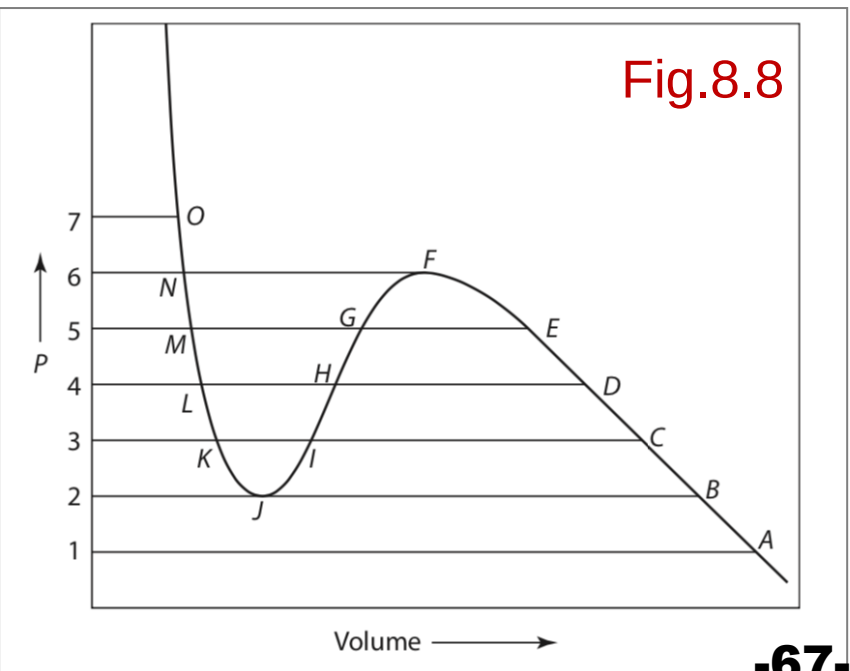
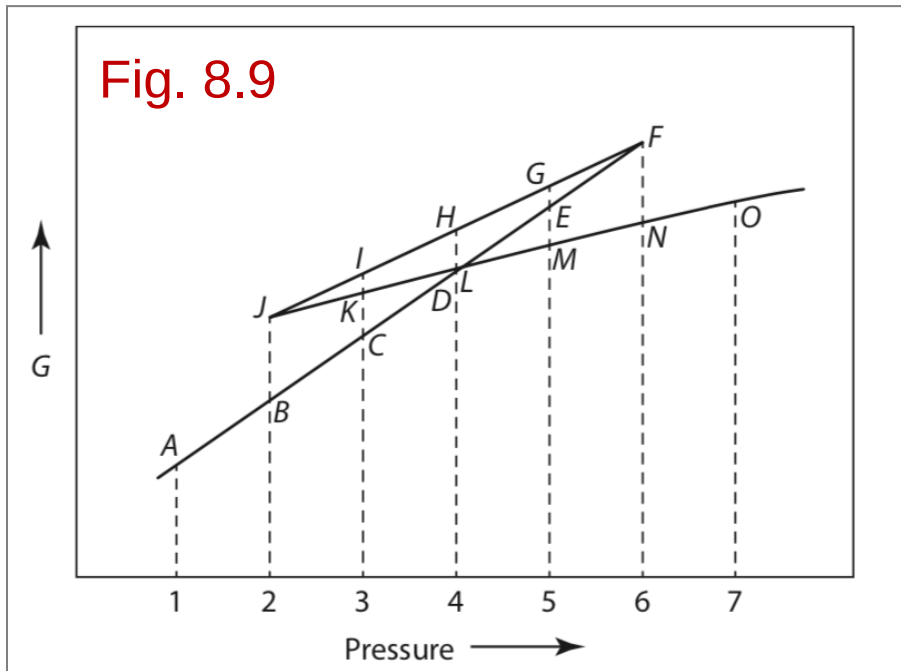


Fig.8.8



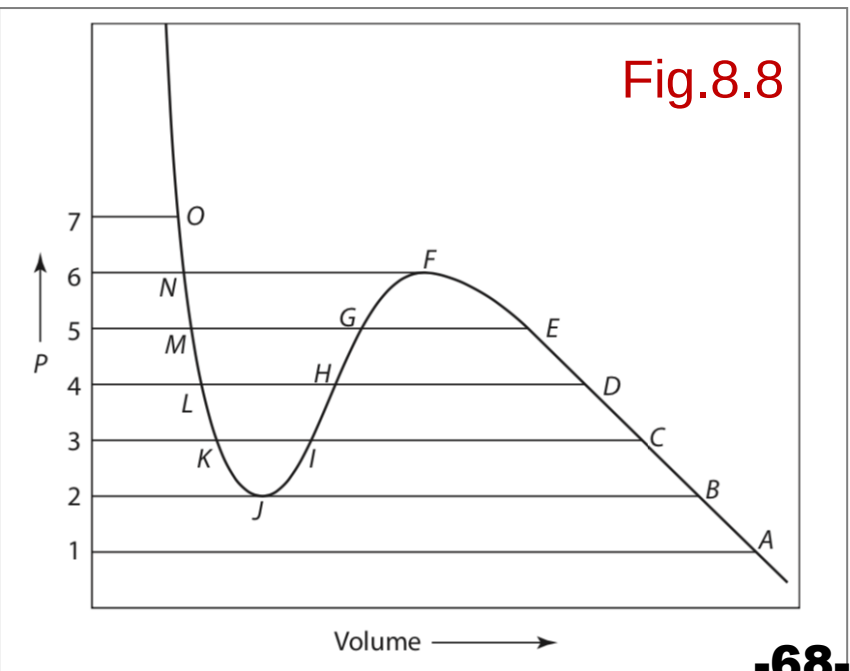
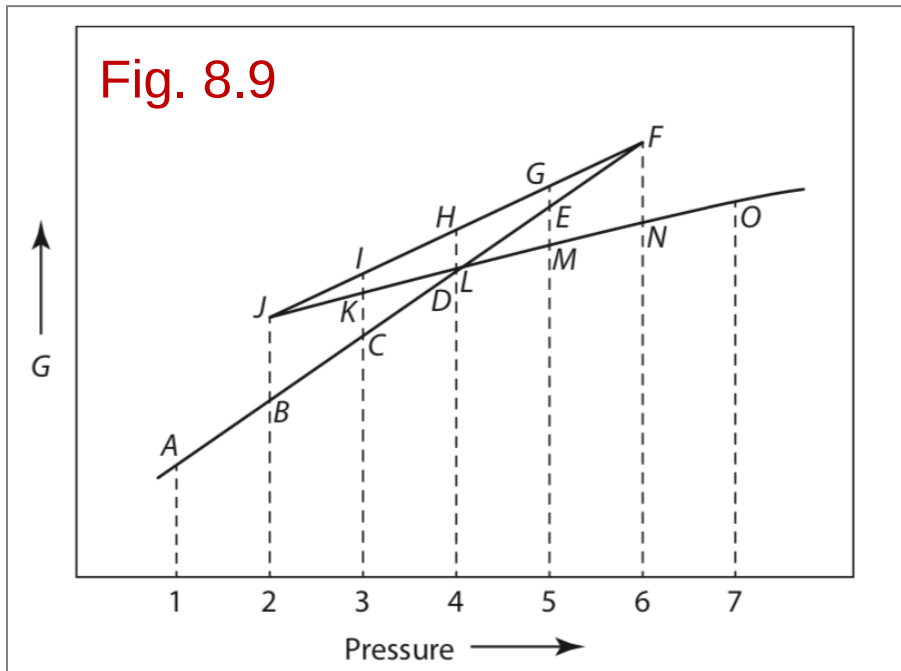
8.5 The Van Der Waals Fluid

- ✓ At P_4 , G_D , which is the molar Gibbs free energy of the vapor phase, equals G_L , which is the molar Gibbs free energy of the liquid phase, and thus, vapor and liquid coexist in equilibrium with one another at the state P_4 , T . The transition from the vapor phase to the liquid phase is a first-order phase transition. In Figure 8.8, a tie-line connects the points D and L across a two-phase region.



8.5 The Van Der Waals Fluid

- ✓ The line **DF** represents the **metastable vapor** phase and the line **LJ** represents the **metastable liquid** states. Thus, in the absence of the formation of the liquid phase from the vapor phase at the state *D*, a **supersaturated vapor** would exist along the curve **DEF**, and, in the absence of the formation of the vapor phase from the liquid phase at the state *L*, **supersaturated liquid** would exist along the line **LKJ**.



8.5 The Van Der Waals Fluid

- ✓ In view of the **violation** of the criterion for intrinsic **stability** over the states path **JHF**, the states represented by this line are unstable, and a **transformation will occur spontaneously** (i.e., without an energy barrier) to a two-phase equilibrium.
- ✓ The points **F** and **J** represent the limits of **metastability** of the vapor and liquid phases, respectively. These were called **spina** by van der Waals (1908).

Fig. 8.9

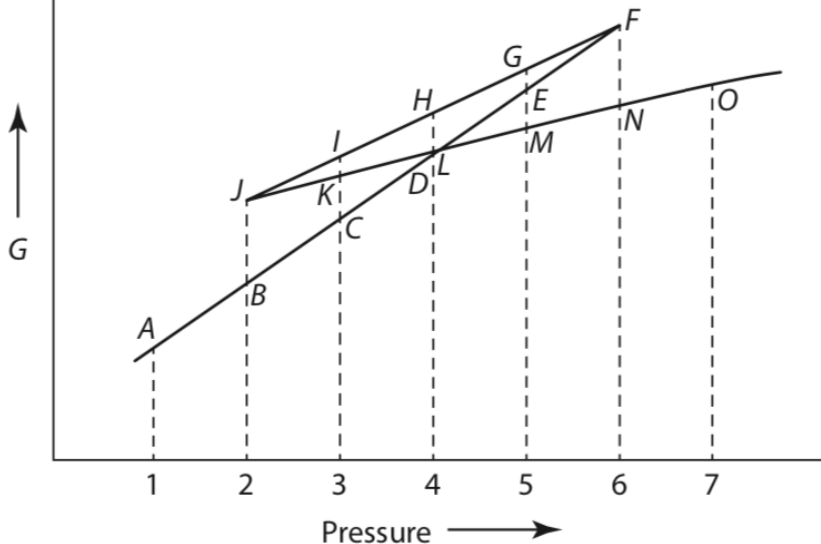
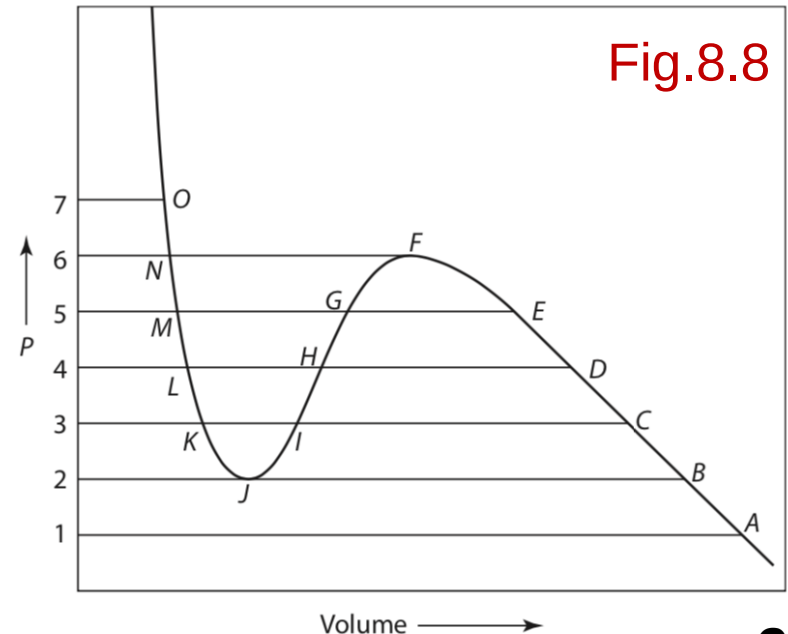


Fig.8.8



8.5 The Van Der Waals Fluid

- ✓ The spina or spinodes occur where the value of $(\partial P/\partial V)_T = 0$, which, for an isotherm of the van der Waals fluid, yields two points. Between **JHGF**, a single-phase absolute equilibrium does not exist.

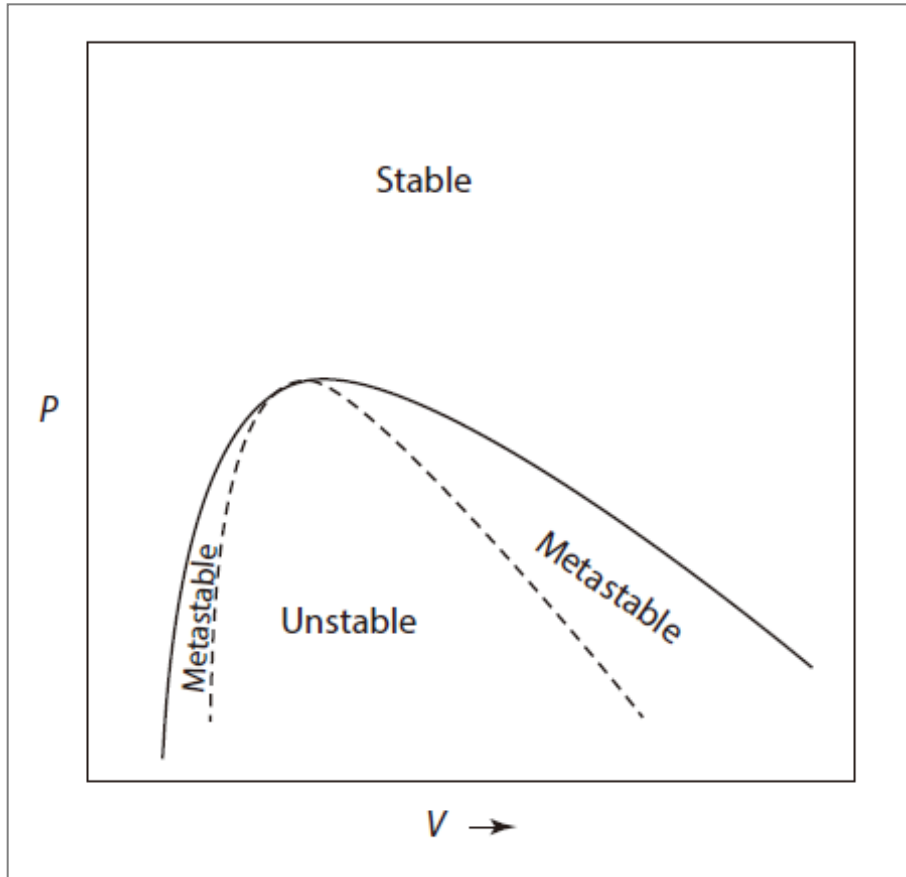
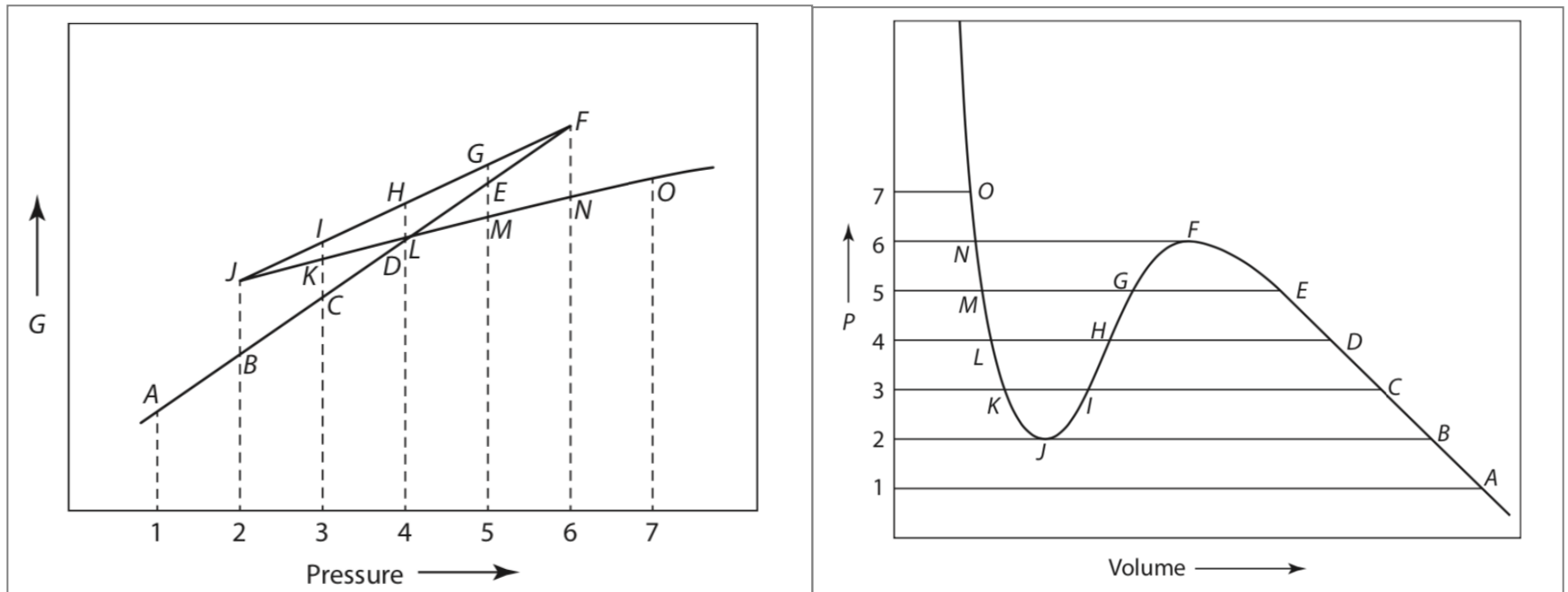


Figure 8.10

P - V diagram showing the locus of spinodal points and regions of metastability and instability.

8.5 The Van Der Waals Fluid

- ✓ It is thus seen that the van der Waals equation predicts the phase change which occurs in the system at temperatures less than T_{cr} . At any temperature below T_{cr} , the value of P for equilibrium between the vapor and liquid phases (e.g., P_4 in Figures 8.8 and 8.9) is that whereby the area HFD equals the area LJH in Figure 8.8. This is called the Maxwell construction.



8.5 The Van Der Waals Fluid

- ✓ The measured values of T_{cr} and P_{cr} for CO_2 are, respectively, 31°C and 72.9 atm .

$$T_{cr} = \frac{8a}{27bR}, \quad V_{cr} = 3b \quad P_{cr} = \frac{a}{27b^2}$$

$$a = 27b^2 P_{cr} = 3.59 \text{ liters}^2 \cdot \text{atm/mole}^2$$

$$b = \frac{RT_{cr}}{8P_{cr}} = 0.0427 \text{ liters/mole}$$

- ✓ Therefore the van der Waals equation for CO_2 is given as

$$\left(P + \frac{3.59}{V^2} \right) (V - 0.0427) = RT$$

8.5 The Van Der Waals Fluid

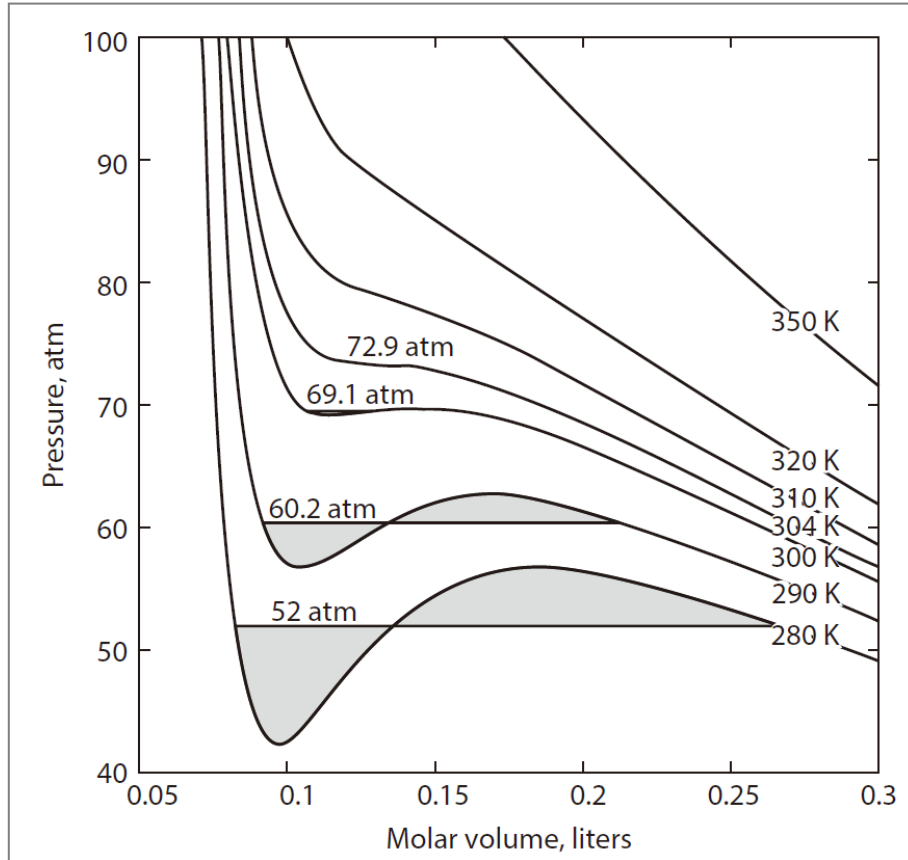


Figure 8.11 shows the variation of P with V for several temperatures. It is seen that the **304 K** isotherm exhibits a horizontal inflection at the critical point. At temperatures lower than 304 K, the isotherms show the expected maxima and minima.

8.5 The Van Der Waals Fluid

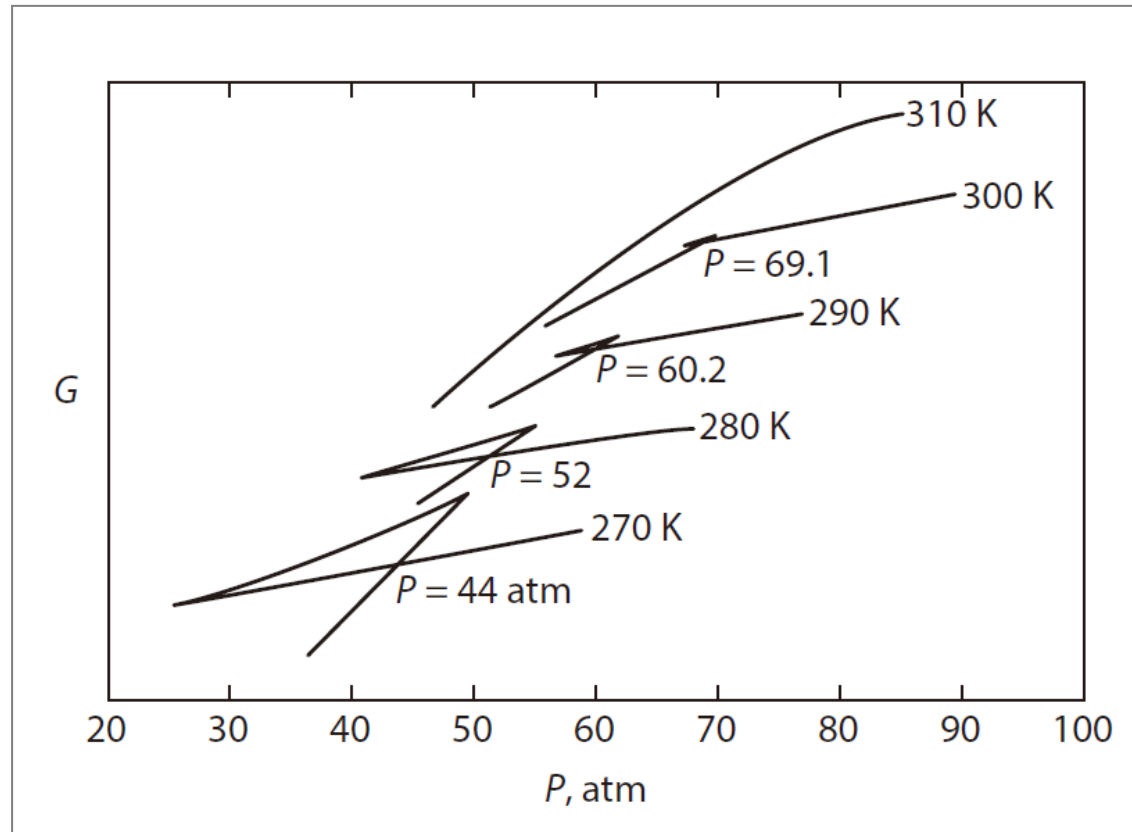
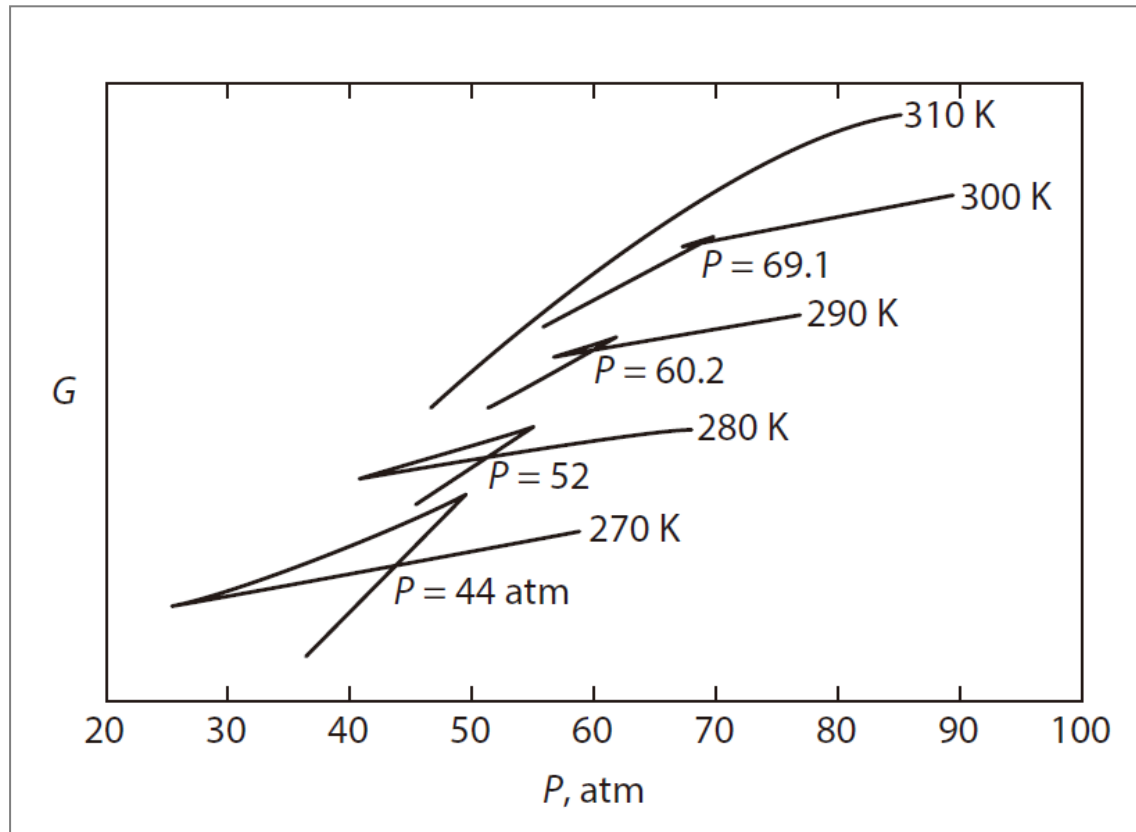


Figure 8.12 shows the variation of the saturated vapor pressure of liquid CO_2 with temperature. Also, as the temperature increases toward the critical point, the range of unstable states diminishes and finally disappears at T_{cr} .

8.5 The Van Der Waals Fluid



At temperatures greater than T_{cr} , the full line indicates that only one phase is stable over the entire range of pressure. Since G_A $\left(G = G_A + \int_{P_A}^P V dP\right)$ is a function of temperature, the positions of the isotherms with respect to one another are arbitrary; only the P -axis is quantitatively significant.

8.5 The Van Der Waals Fluid

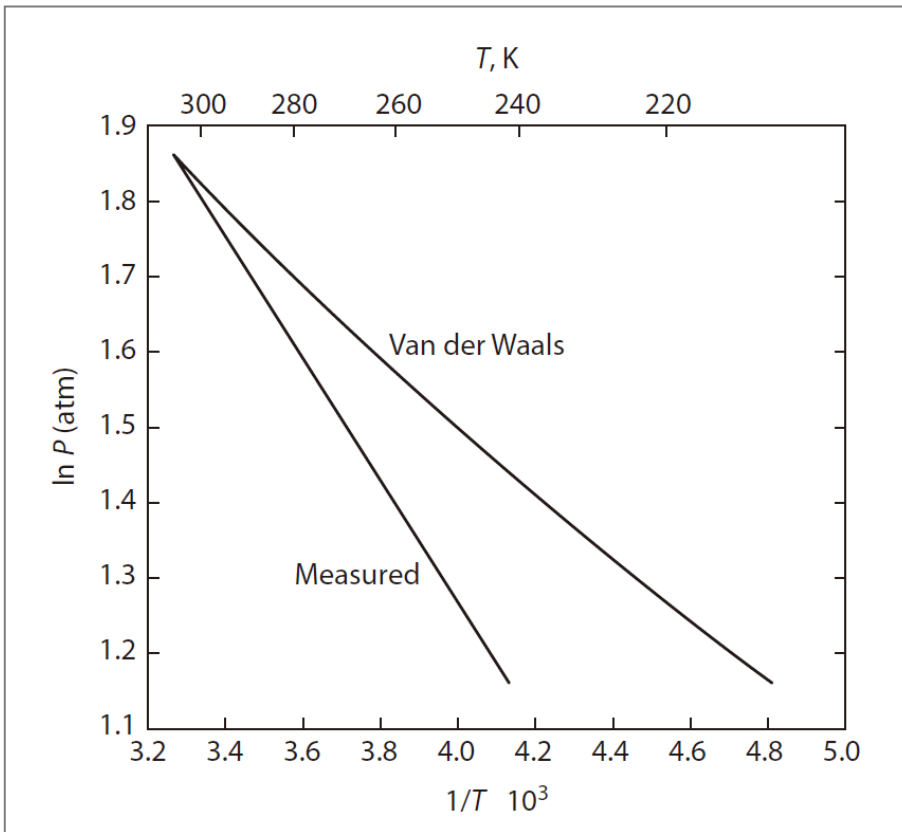


Figure 8.13

Comparison of the variation, with temperature, of the vapor pressure of van der Waals liquid CO_2 with the measured vapor pressures.

Comparison shows that the van der Waals equation predicts values of vapor pressure which are higher than the measured values, although the difference between the two values decreases with increasing temperature.

Consequently, the van der Waals equation predicts a value of the molar latent heat of evaporation of liquid CO_2 which is less than the measured value, with ΔH_{evap} being obtained as $-2.303 R \times$ (the slope of the line).

8.5 The Van Der Waals Fluid

- ✓ The molar latent heat of evaporation of a liquefied van der Waals fluid can be calculated as follows:

$$\Delta H_{\text{evap}} = H_v - H_l = U_v - U_l + P(V_v - V_l)$$

V_v and V_l are, respectively, the molar volumes of the coexisting vapor and liquid phases

P is the saturated vapor pressure at the temperature T

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial P}{\partial T}\right)_V - P$$

$$dU = TdS - PdV$$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$$

When applied to the van der Waals fluid, we have

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{R}{V - b}\right) - P = \frac{a}{V^2}$$

8.5 The Van Der Waals Fluid

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{R}{V-b}\right) - P = \frac{a}{V^2} \Rightarrow U = -\frac{a}{V} - \text{constant}$$

The integration constant is a function of temperature

$$\begin{aligned}\Delta H_{\text{evap}} &= -\frac{a}{V_v} + \frac{a}{V_l} + P(V_v - V_l) \\ &= -a\left(\frac{1}{V_v} - \frac{1}{V_l}\right) + \frac{a}{V_l} + P(V_v - V_l)\end{aligned}$$

This equation correctly predicts that ΔH_{evap} for a van der Waals fluid rapidly falls to zero as the temperature approaches T_{cr} in which state, $V_v = V_l$.

8.5 The Van Der Waals Fluid

- ✓ Although van der Waals developed his equation from a consideration of the physical factors causing nonideal behavior, the requirement that the pressure, volume, and temperature at the critical point be known for the calculation of a and b means that the equation is empirical.
- ✓ This, however, does not detract from the usefulness of the equation in representing the behavior of a gas which exhibits a relatively small departure from ideality.
- ✓ Importantly, the equation of state predicts a change of phase which is first order (when the vapor transforms in the liquid). It also predicts a continuous change of state when above the critical values of P and T .

Homework

1. Demonstrate the law of corresponding states by writing the van der Waals equation in terms of the reduced variables. Calculate the compressibility factor for a van der Waals fluid at its critical point and compare the result with the values obtained for real gases at their critical points listed in Table 8.1. Calculate the value of $(\partial U/\partial V)_T$ for a van der Waals fluid.
2. The virial equation of state for n-butane at 460 K is $Z = 1 + A/V + B/V^2$, in which $A = -265 \text{ cm}^3/\text{g}\cdot\text{mole}$ and $B = 30250 \text{ cm}^6/\text{g}\cdot\text{mole}^2$. Calculate the work required to reversibly compress 1 mole of n-butane from 50 to 100 atm at 460 K.

Homework

3. Using the virial equation of state for hydrogen at 298 K given in Problem 8.6, calculate
- a. The fugacity of hydrogen at 500 atm and 298 K
 - b. The pressure at which the fugacity is twice the pressure
 - c. The change in the Gibbs free energy caused by a compression of 1 mole of hydrogen at 298 K from 1 to 500 atm
- What is the magnitude of the contribution to (c) caused by the nonideality of hydrogen?

Thermodynamics of Materials & Phase transformations

The End

